GRADE 7 ACCELERATED · UNIT 6

Relationships in Triangles

Eight lessons in teaching order · the next slide jumps to each one

Contents

6.01–02	Triangle Inequality Theorem; Converse of the Pythagorean Theorem	slide 3
6.03	Pythagorean Theorem	slide 38
6.04	Pythagorean Theorem in Real-World Problems	slide 73
6.05	Distance on the Coordinate Plane	slide 108
6.06	Real-World Coordinate-Plane Problems; Perimeter	slide 143
6.07–08	Similar Triangles; Indirect Measurement	slide 178
T-B1	Angle Pairs: Supplementary, Complementary, Vertical, Adjacent	slide 213
T-B2	Interior and Exterior Angles of a Triangle	slide 248

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 1–2

Triangle Inequality Theorem; Converse of the Pythagorean Theorem

MA.8.GR.1.3

I can decide whether three lengths can make a triangle, and whether they make a right triangle.

Warm-Up

1. A car gets 32 miles per gallon. How far does it go on 9 gallons?
2. A figure of area $5\,\text{cm}^2$ is scaled by $k = 3$. The copy's area?
3. Find the value: $\sqrt{169}$
4. Find the value: $5^2 + 12^2$

Warm-Up

1. A car gets 32 miles per gallon. How far does it go on 9 gallons?

   288 mi

2. A figure of area $5\,\text{cm}^2$ is scaled by $k = 3$. The copy's area?

   45 cm²

3. Find the value: $\sqrt{169}$ **13**

4. Find the value: $5^2 + 12^2$ **169**

Vocabulary

triangle inequality

any two sides of a triangle add up to MORE than the third side

legs

the two sides of a right triangle that make the right angle

hypotenuse

the side across from the right angle — always the longest side

converse

a statement turned around: if $a^2 + b^2 = c^2$ for the two shorter sides and the longest, the triangle is right

Notes

I. Two sides together must be longer than the third



3, 4, 9
a gap

4, 5, 9
flat

5, 6, 9
a triangle

Add the two shorter sides. A triangle needs MORE than the longest side.

On the FAST sheet: $a + b > c$, $a + c > b$, $b + c > a$

Notes

II. The third side is between the difference and the sum

$$9 - 4 = 5 \qquad 9 + 4 = 13$$

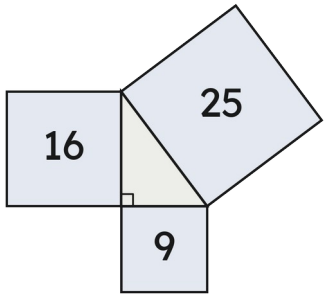
$$5 < x < 13$$

Sides 4 and 9: the third side x is more than 5 and less than 13.

Never $\leq$: at 5 or 13 the sides lie flat.

Notes

III. The legs' squares add up to the hypotenuse's square



$$3^2 + 4^2 = 9 + 16 = 25 = 5^2$$

Two shorter sides squared and added = the longest side squared: a right triangle.

9, 15, 12: $9^2 + 12^2 = 225 = 15^2$, right. 5, 6, 8: $5^2 + 6^2 = 61$, not 64.

Example 1

Can sides of 5 cm, 8 cm and 14 cm make a triangle?

Example 1

$5 + 8 = 13$

$13 < 14$

Answer: No — 5 + 8 = 13 is less than 14.

Your Turn

Can sides of 7 inches, 9 inches and 15 inches make a triangle?

Your Turn

Can sides of 7 inches, 9 inches and 15 inches make a triangle?

$7 + 9 = 16$

16 is more than 15: a triangle.

Answer: Yes — 7 + 9 = 16 is more than 15.

Example 2

A triangle has sides of 20 m, 12 m and 16 m.

Is it a right triangle?

Example 2

$12^2 + 16^2 = 144 + 256 = 400$

$20^2 = 400$

Answer: Yes — $12^2 + 16^2 = 20^2$.

Your Turn

A triangle has sides of 8 ft, 11 ft and 14 ft.

Is it a right triangle?

Your Turn

A triangle has sides of 8 ft, 11 ft and 14 ft.

Is it a right triangle?

$8^2 + 11^2 = 64 + 121 = 185$

$14^2 = 196$: not equal.

Answer: No — 185 is not 196.

Whiteboards · Question 1 of 9

Can sides of 2 ft, 7 ft and 10 ft make a triangle?

Answer it.

Whiteboards · Question 1 of 9

Can sides of 2 ft, 7 ft and 10 ft make a triangle?

$2 + 7 = 9$

9 is less than 10: no triangle.

Answer: No — 2 + 7 = 9 is less than 10.

Whiteboards · Question 2 of 9

Can sides of 6 m, 6 m and 11 m make a triangle?

Answer it.

Whiteboards · Question 2 of 9

Can sides of 6 m, 6 m and 11 m make a triangle?

$6 + 6 = 12$

12 is more than 11: a triangle.

Answer: Yes — 6 + 6 = 12 is more than 11.

Whiteboards · Question 3 of 9

Which three lengths can make a triangle?

A. 7, 7, 14

B. 5, 11, 8

C. 3, 10, 6

D. 12, 4, 6

Answer it.

Whiteboards · Question 3 of 9

Which three lengths can make a triangle?

$5 + 8 = 13$, more than 11: a triangle.

In each other set the two shorter sides fall short or lie flat.

Answer: B — 5, 11, 8

Whiteboards · Question 4 of 9

Two sides of a triangle are 7 cm and 12 cm.

Write an inequality for the length x, in centimeters, of the third side.

Answer it.

Whiteboards · Question 4 of 9

Two sides of a triangle are 7 cm and 12 cm.

Write an inequality for the length x, in centimeters, of the third side.

$$12 - 7 = 5 \quad \text{and} \quad 12 + 7 = 19$$

$$5 < x < 19$$

Answer: $5 < x < 19$ — more than 5 cm and less than 19 cm.

Whiteboards · Question 5 of 9

Two sides of a triangle are 6 cm and 10 cm.

Which inequality shows every possible length x, in centimeters, of the third side?

A. $6 < x < 10$

B. $4 \leq x \leq 16$

C. $4 < x < 16$

D. $0 < x < 16$

Answer it.

Whiteboards · Question 5 of 9

Two sides of a triangle are 6 cm and 10 cm.

Which inequality shows every possible length x, in centimeters, of the third side?

$10 - 6 = 4$ and $10 + 6 = 16$

$4 < x < 16$ — never equal

Answer: C — $4 < x < 16$

Whiteboards · Question 6 of 9

Do sides of 8 inches, 15 inches and 17 inches make a right triangle?

Answer it.

Whiteboards · Question 6 of 9

Do sides of 8 inches, 15 inches and 17 inches make a right triangle?

$8^2 + 15^2 = 64 + 225 = 289$

$17^2 = 289$: equal — a right triangle.

Answer: Yes — $8^2 + 15^2 = 289 = 17^2$.

Whiteboards · Question 7 of 9

A triangle has sides of 25 cm, 7 cm and 24 cm.

Is it a right triangle? Which answer has the right reason?

A. Yes, because $7^2 + 24^2 = 25^2$

B. Yes, because $7 + 24 > 25$

C. No, because $25^2 + 7^2 \neq 24^2$

D. No, because $7 + 24 \neq 25$

Answer it.

Whiteboards · Question 7 of 9

A triangle has sides of 25 cm, 7 cm and 24 cm.

Is it a right triangle? Which answer has the right reason?

$$7^2 + 24^2 = 49 + 576 = 625$$

$25^2 = 625$: equal — right.

Answer: A — yes, because $7^2 + 24^2 = 25^2$

Whiteboards · Question 8 of 9

A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds.

Does the sail have a right angle?

Answer it.

Whiteboards · Question 8 of 9

A triangular sail has sides of 20 ft, 21 ft and 29 ft, and it weighs 12 pounds.

Does the sail have a right angle?

$$20^2 + 21^2 = 400 + 441 = 841$$

$29^2 = 841$: a right angle.

Answer: Yes — $20^2 + 21^2 = 841 = 29^2$.

Whiteboards · Question 9 of 9

Ana says sides of 6, 7 and 9 make a right triangle because $6 + 7 > 9$.

Is she right? Explain.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Ana says sides of 6, 7 and 9 make a right triangle because 6 + 7 > 9.

Is she right? Explain.

$6^2 + 7^2 = 85$, but $9^2 = 81$.

Answer: No — 6 + 7 > 9 shows only a triangle; 85 is not 81, so it is not right.

Independent Set

1. Can sides of 4 cm, 8 cm and 13 cm make a triangle?
2. Can sides of 9 inches, 9 inches and 18 inches make a triangle?
3. Two sides of a triangle are 5 m and 11 m. Write an inequality for the length x, in meters, of the third side.
4. Do sides of 28 cm, 45 cm and 53 cm make a right triangle?
5. Do sides of 17 ft, 10 ft and 14 ft make a right triangle?
6. A carpenter joins boards of 30 inches, 40 inches and 50 inches into a triangle. Is one corner a right angle?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Triangle inequality — 2N5

Converse of the Pythagorean theorem: is it a right triangle? — EQZ

GRADE 7 ACCELERATED · UNIT 6 · LESSON 3

Pythagorean Theorem

MA.8.GR.1.1

I can find a missing side of a right triangle — the hypotenuse, or a leg.

Warm-Up

1. Can sides of 4 inches, 7 inches and 12 inches make a triangle?
2. A printer prints 18 pages a minute. Pages in 4.5 minutes?
3. Find the value: $\sqrt{196}$
4. Find the value: $10^2 - 6^2$

Warm-Up

1. Can sides of 4 inches, 7 inches and 12 inches make a triangle?

   No — 4 + 7 = 11 < 12

2. A printer prints 18 pages a minute. Pages in 4.5 minutes?

   81 pages

3. Find the value: $\sqrt{196}$ **14**

4. Find the value: $10^2 - 6^2$ **64**

Vocabulary

Pythagorean Theorem

in every right triangle, leg² + leg² = hypotenuse²: $a^2 + b^2 = c^2$

hypotenuse

c — the side across from the right angle, always the longest side

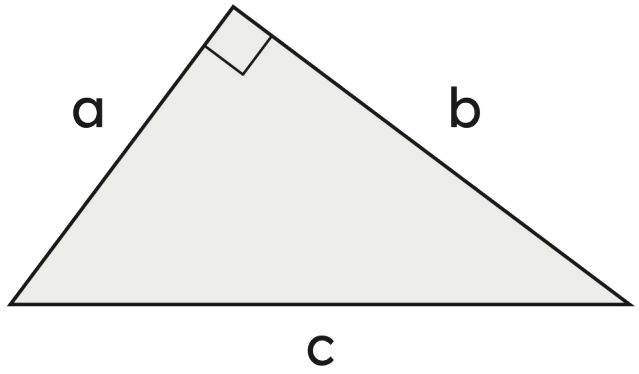
square root

the number that, times itself,

makes the number under the root sign: $\sqrt{64} = 8$

Notes

I. The square corner first; c is across from it



$$a^2 + b^2 = c^2$$

The legs' squares add up to the hypotenuse's square. c is across from the right angle — always the longest side.

Notes

II. The hypotenuse: square, add, then the root

$$6^2 + 8^2 = 36 + 64 = 100$$

$$c = \sqrt{100} = 10$$

Legs 3 and 5: $3^2 + 5^2 = 34$, $c = \sqrt{34} \approx 5.8$

Exact: $\sqrt{34}$. To the nearest tenth: 5.8.

Notes

III. A leg: subtract the squares

$$a^2 + 15^2 = 17^2$$

$$a^2 = 289 - 225 = 64$$

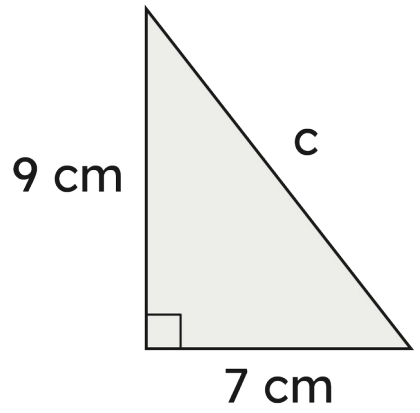
$$a = \sqrt{64} = 8$$

For a leg, subtract: $a^2 = c^2 - b^2$.

Check: a leg is shorter than the hypotenuse.

Example 1

A right triangle has legs of 7 cm and 9 cm.



How long is the hypotenuse? Give it exactly and to the nearest tenth.

Example 1

$$c^2 = 7^2 + 9^2 = 49 + 81 = 130$$

$$c = \sqrt{130} \approx 11.4$$

Answer: $c = \sqrt{130}$ cm ≈ 11.4 cm

Your Turn

A right triangle has legs of 5 inches and 11 inches.

How long is the hypotenuse, exactly and to the nearest tenth?

Your Turn

A right triangle has legs of 5 inches and 11 inches.

How long is the hypotenuse, exactly and to the nearest tenth?

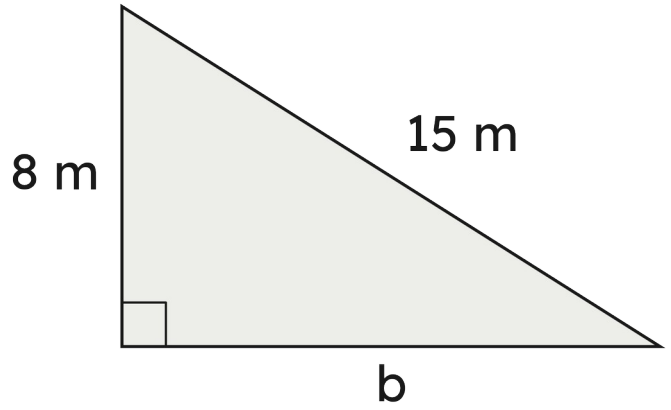
$$c^2 = 5^2 + 11^2 = 25 + 121 = 146$$

$$c = \sqrt{146} \approx 12.1$$

Answer: $c = \sqrt{146}$ in ≈ 12.1 in

Example 2

A right triangle has a hypotenuse of 15 m and one leg of 8 m.



How long is the other leg? Give it exactly and to the nearest tenth.

Example 2

$$b^2 + 8^2 = 15^2$$

$$b^2 = 225 - 64 = 161$$

$$b = \sqrt{161} \approx 12.7$$

Answer: $b = \sqrt{161}$ m ≈ 12.7 m

Your Turn

A right triangle has a hypotenuse of 11 ft and one leg of 6 ft.

How long is the other leg, exactly and to the nearest tenth?

Your Turn

A right triangle has a hypotenuse of 11 ft and one leg of 6 ft.

How long is the other leg, exactly and to the nearest tenth?

$$b^2 = 11^2 - 6^2 = 121 - 36 = 85$$

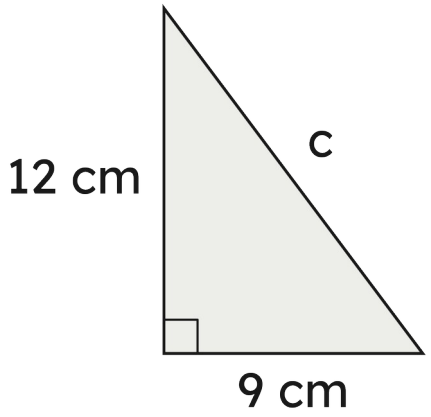
$$b = \sqrt{85} \approx 9.2$$

Answer: $b = \sqrt{85}$ ft ≈ 9.2 ft

Whiteboards · Question 1 of 9

A right triangle has legs of 9 cm and 12 cm.

How long is the hypotenuse?

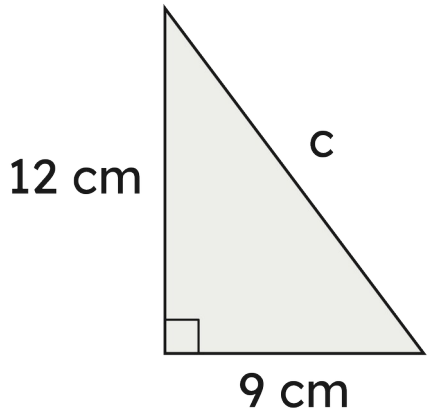


Answer it.

Whiteboards · Question 1 of 9

A right triangle has legs of 9 cm and 12 cm.

How long is the hypotenuse?



$c^2 = 81 + 144 = 225$

$c = \sqrt{225} = 15$

Answer: 15 cm

Whiteboards · Question 2 of 9

A right triangle has legs of 4 inches and 9 inches.

How long is the hypotenuse, to the nearest tenth?

Answer it.

Whiteboards · Question 2 of 9

A right triangle has legs of 4 inches and 9 inches.

How long is the hypotenuse, to the nearest tenth?

$$c^2 = 16 + 81 = 97$$

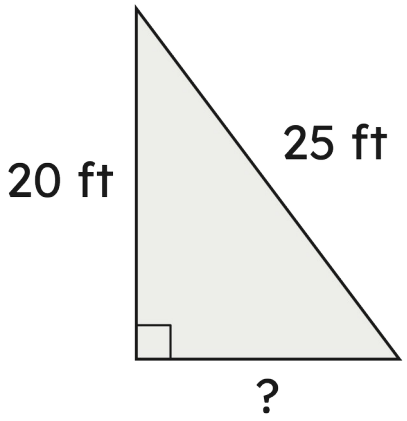
$$c = \sqrt{97} \approx 9.8$$

Answer: $\sqrt{97}$ in ≈ 9.8 in

Whiteboards · Question 3 of 9

A right triangle has a hypotenuse of 25 ft and one leg of 20 ft.

How long is the other leg?

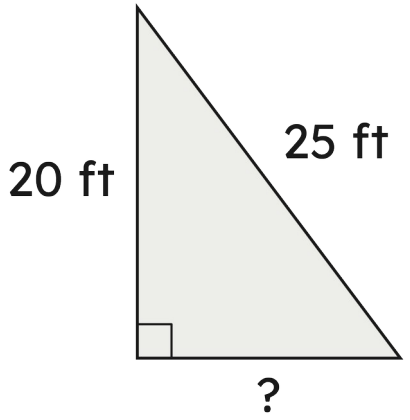


Answer it.

Whiteboards · Question 3 of 9

A right triangle has a hypotenuse of 25 ft and one leg of 20 ft.

How long is the other leg?



$$b^2 = 625 - 400 = 225$$

$$b = \sqrt{225} = 15$$

Answer: 15 ft

Whiteboards · Question 4 of 9

A right triangle has a hypotenuse of 41 m and one leg of 40 m.

How long is the other leg?

A. 9 m

B. 81 m

C. 1 m

D. 57.3 m

Answer it.

Whiteboards · Question 4 of 9

A right triangle has a hypotenuse of 41 m and one leg of 40 m.

How long is the other leg?

$$b^2 = 1{,}681 - 1{,}600 = 81$$

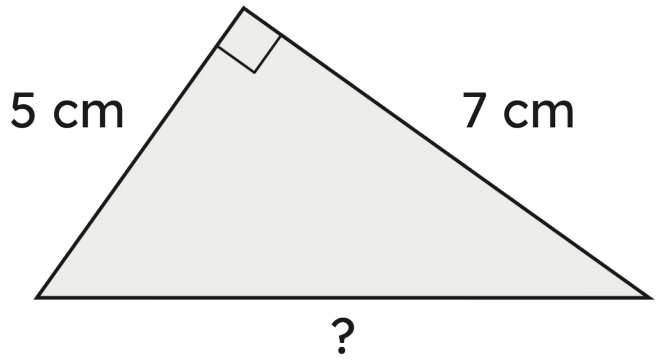
$$b = \sqrt{81} = 9$$

Answer: A — 9 m

Whiteboards · Question 5 of 9

The right triangle is drawn turned. Its legs are 5 cm and 7 cm.

How long is the hypotenuse, to the nearest tenth?

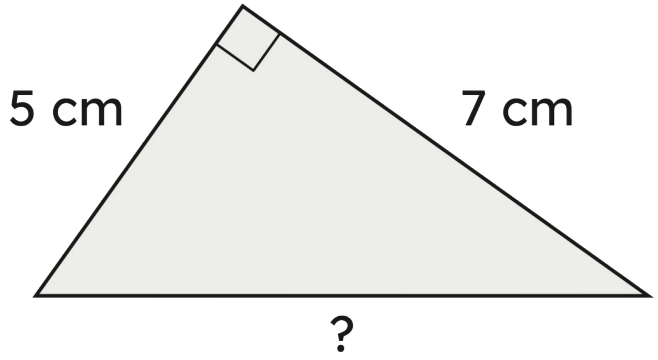


Answer it.

Whiteboards · Question 5 of 9

The right triangle is drawn turned. Its legs are 5 cm and 7 cm.

How long is the hypotenuse, to the nearest tenth?



$c^2 = 25 + 49 = 74$

$c = \sqrt{74} \approx 8.6$

Answer: $\sqrt{74}$ cm ≈ 8.6 cm

Whiteboards · Question 6 of 9

A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x.

Which equation finds x?

A. $16 + x = 20$

B. $16^2 + 20^2 = x^2$

C. $16^2 + x^2 = 2 \cdot 20$

D. $16^2 + x^2 = 20^2$

Answer it.

Whiteboards · Question 6 of 9

A right triangle's hypotenuse is 20 inches; its legs are 16 inches and x.

Which equation finds x?

$$x^2 = 400 - 256 = 144$$

$$x = \sqrt{144} = 12$$

Answer: D — and x = 12 in

Whiteboards · Question 7 of 9

A right triangle has a hypotenuse of 14 m and one leg of 9 m.

How long is the other leg, to the nearest tenth?

Answer it.

Whiteboards · Question 7 of 9

A right triangle has a hypotenuse of 14 m and one leg of 9 m.

How long is the other leg, to the nearest tenth?

$$b^2 = 196 - 81 = 115$$

$$b = \sqrt{115} \approx 10.7$$

Answer: $\sqrt{115}$ m ≈ 10.7 m

Whiteboards · Question 8 of 9

A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars.

How long is its longest side, to the nearest tenth?

Answer it.

Whiteboards · Question 8 of 9

A pennant is cut as a right triangle with legs of 8 inches and 11 inches. The fabric costs 6 dollars.

How long is its longest side, to the nearest tenth?

$$c^2 = 64 + 121 = 185$$

$$c = \sqrt{185} \approx 13.6$$

Answer: $\sqrt{185}$ in $\approx$ 13.6 in

Whiteboards · Question 9 of 9

Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm.

She writes $\sqrt{5^2 + 13^2} \approx 13.9$.

What went wrong? How long is the leg?

Write your answer in sentences.

Whiteboards · Question 9 of 9

Jo finds a leg of a right triangle. The hypotenuse is 13 cm and the other leg is 5 cm.

She writes $\sqrt{5^2 + 13^2} \approx 13.9$.

What went wrong? How long is the leg?

$$b^2 = 169 - 25 = 144$$

$$b = \sqrt{144} = 12$$

Answer: She added the squares; a leg must be shorter than 13 cm. Subtract: the leg is 12 cm.

Independent Set

1. A right triangle has legs of 12 m and 5 m. How long is the hypotenuse?

2. A right triangle has legs of 3 ft and 8 ft. How long is the hypotenuse, exactly and to the nearest tenth?

3. A right triangle has a hypotenuse of 39 cm and one leg of 36 cm. How long is the other leg?

4. A right triangle has a hypotenuse of 11 inches and one leg of 4 inches. How long is the other leg, exactly and to the nearest tenth?

5. A right triangle has legs of 6 m and 11 m (drawn turned on its side on the paper). How long is its hypotenuse, to the nearest tenth?

6. A right triangle has a hypotenuse of 25 yd and one leg of 24 yd. The other leg is x. Write the equation that finds x, then find x.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Pythagorean theorem: find the missing leg or hypotenuse length — MTM

GRADE 7 ACCELERATED · UNIT 6 · LESSON 4

Pythagorean Theorem in Real-World Problems

MA.8.GR.1.1

I can find a length in a real situation by finding the right triangle inside it.

Warm-Up

1. A right triangle: hypotenuse 25 ft, one leg 24 ft. The other leg?
2. Do sides of 9, 40 and 41 make a right triangle?
3. A plan's scale is 1 in : 4 ft. A wall is 3.5 inches on it. Real length?
4. How many right angles does a rectangle have?

Warm-Up

1. A right triangle: hypotenuse 25 ft, one leg 24 ft. The other leg?

   7 ft

2. Do sides of 9, 40 and 41 make a right triangle?

   Yes — $81 + 1{,}600 = 1{,}681 = 41^2$

3. A plan's scale is 1 in : 4 ft. A wall is 3.5 inches on it. Real length?

   14 ft

4. How many right angles does a rectangle have? **4**

Vocabulary

diagonal

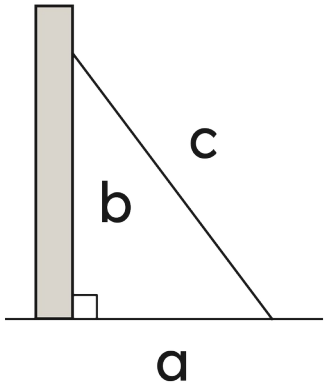
the segment from one corner of a rectangle to the opposite corner — the hypotenuse of the right triangle it cuts off

rise and run

a ramp's height and the distance along the ground it covers; the ramp itself is the hypotenuse

Notes

I. Find the right angle, then c

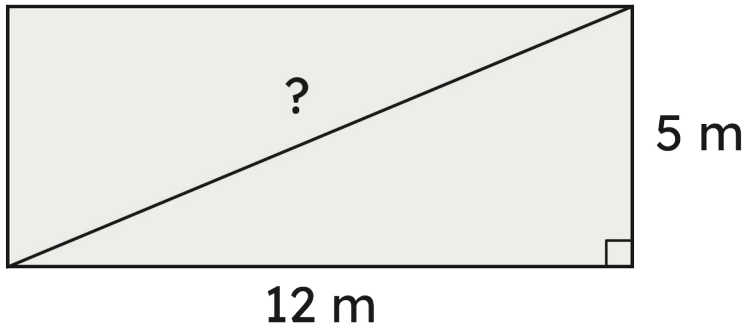


Sketch the story. Mark the right angle. The side across from it is c.

A ladder, a diagonal, a ramp, a kite string: each is the hypotenuse.

Notes

II. c in a story: the diagonal



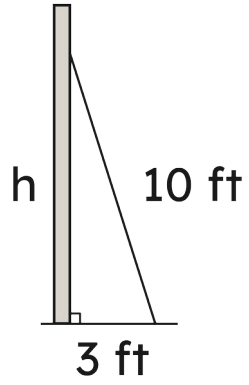
$$c^2 = 5^2 + 12^2 = 25 + 144 = 169$$

$$c = \sqrt{169} = 13$$

The path is 13 m. Along two sides is 17 m: the path is 4 m shorter.

Notes

III. A leg in a story: how high

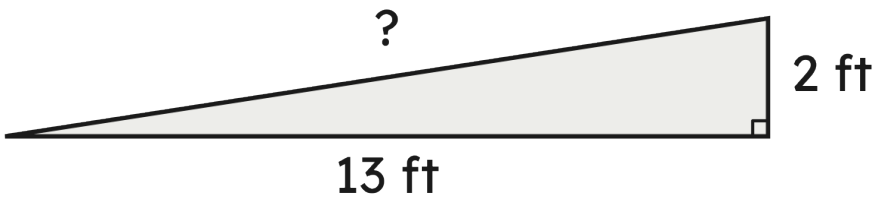


$$h^2 = 10^2 - 3^2 = 100 - 9 = 91$$

$$h = \sqrt{91} \approx 9.5$$

Example 1

A ramp rises 2 ft over a run of 13 ft along the ground.



How long is the ramp, to the nearest tenth of a foot?

Example 1

$$c^2 = 2^2 + 13^2 = 4 + 169 = 173$$

$$c = \sqrt{173} \approx 13.2$$

Answer: The ramp is about 13.2 ft long.

Your Turn

A computer screen is 7 inches tall and 10 inches wide.

How long is its diagonal, to the nearest tenth of an inch?

Your Turn

A computer screen is 7 inches tall and 10 inches wide.

How long is its diagonal, to the nearest tenth of an inch?

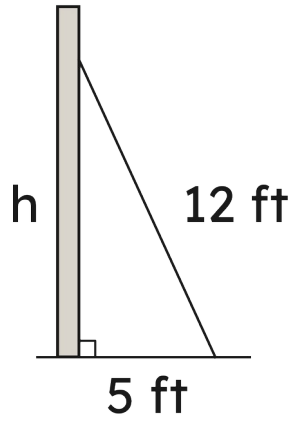
$$c^2 = 7^2 + 10^2 = 49 + 100 = 149$$

$$c = \sqrt{149} \approx 12.2$$

Answer: About 12.2 in

Example 2

A 12-ft ladder leans on a wall. Its foot is 5 ft from the wall.



How high up the wall does it reach, to the nearest tenth of a foot?

Example 2

$$h^2 = 12^2 - 5^2 = 144 - 25 = 119$$

$$h = \sqrt{119} \approx 10.9$$

Answer: It reaches about 10.9 ft up the wall.

Your Turn

A 13-ft ladder leans on a wall with its foot 6 ft from the wall.

How high does it reach, to the nearest tenth of a foot?

Your Turn

A 13-ft ladder leans on a wall with its foot 6 ft from the wall.

How high does it reach, to the nearest tenth of a foot?

$$h^2 = 13^2 - 6^2 = 169 - 36 = 133$$

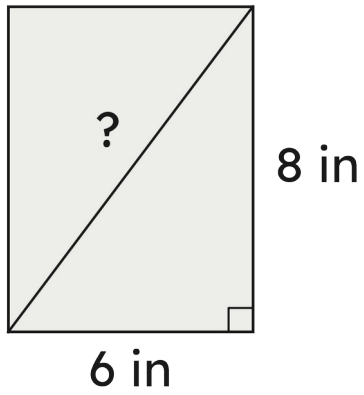
$$h = \sqrt{133} \approx 11.5$$

Answer: About 11.5 ft

Whiteboards · Question 1 of 9

A tablet screen is 6 inches wide and 8 inches tall.

How long is its diagonal?

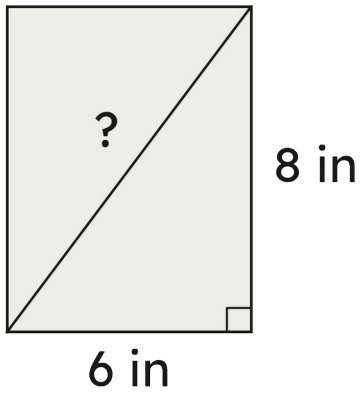


Answer it.

Whiteboards · Question 1 of 9

A tablet screen is 6 inches wide and 8 inches tall.

How long is its diagonal?



$$c^2 = 36 + 64 = 100$$

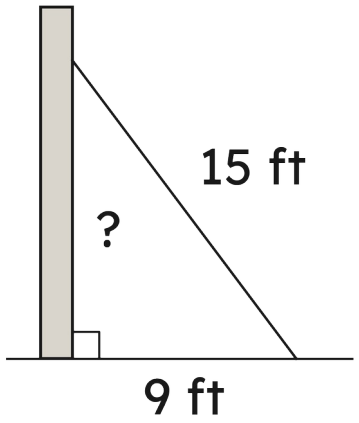
$$c = \sqrt{100} = 10$$

Answer: 10 in

Whiteboards · Question 2 of 9

A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall.

How high up the wall does it reach?

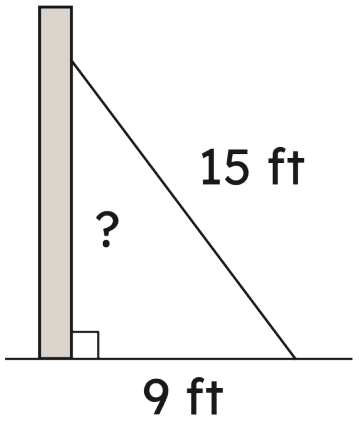


Answer it.

Whiteboards · Question 2 of 9

A 15-ft ladder leans on a wall. Its foot is 9 ft from the wall.

How high up the wall does it reach?



$h^2 = 225 - 81 = 144$

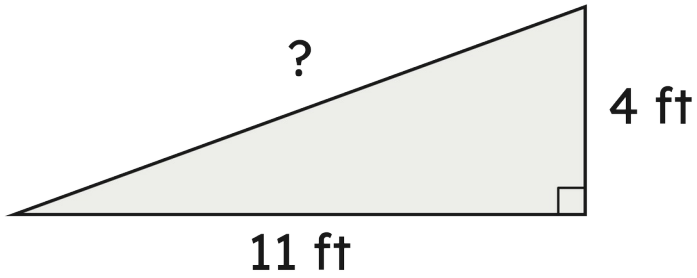
$h = \sqrt{144} = 12$

Answer: 12 ft

Whiteboards · Question 3 of 9

A ramp rises 4 ft over a run of 11 ft.

How long is the ramp, to the nearest tenth of a foot?

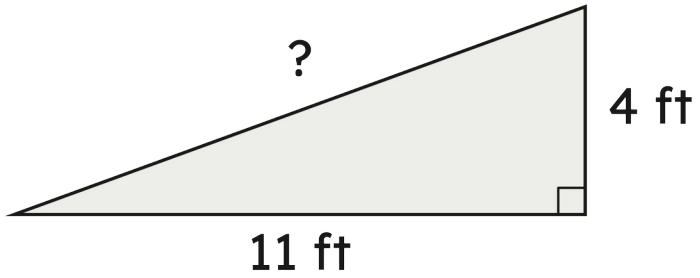


Answer it.

Whiteboards · Question 3 of 9

A ramp rises 4 ft over a run of 11 ft.

How long is the ramp, to the nearest tenth of a foot?



$c^2 = 16 + 121 = 137$

$c = \sqrt{137} \approx 11.7$

Answer: About 11.7 ft

Whiteboards · Question 4 of 9

A 20-ft ladder reaches 16 ft up a wall.

How far from the wall is its foot?

A. 12 ft

B. 25.6 ft

C. 4 ft

D. 144 ft

Answer it.

Whiteboards · Question 4 of 9

A 20-ft ladder reaches 16 ft up a wall.

How far from the wall is its foot?

$$a^2 = 400 - 256 = 144$$

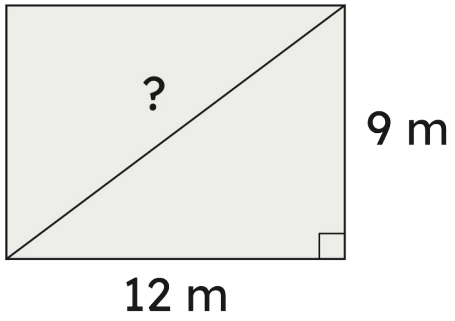
$$a = \sqrt{144} = 12$$

Answer: A — 12 ft

Whiteboards · Question 5 of 9

A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner.

How much shorter is Ava's walk?

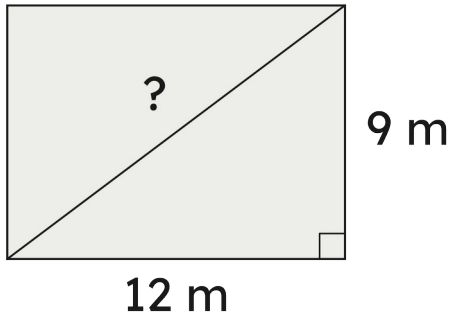


Answer it.

Whiteboards · Question 5 of 9

A lawn is 9 m by 12 m. Ben walks along two sides; Ava walks straight across, corner to corner.

How much shorter is Ava's walk?



Along two sides: $9 + 12 = 21$

Across: $\sqrt{81 + 144} = 15$

$21 - 15 = 6$

Answer: 6 m shorter

Whiteboards · Question 6 of 9

A kite flies on 30 m of string, straight above a spot 18 m from where the string is held.

Which length is the hypotenuse?

A. the 18 m on the ground

B. the kite's height

C. the 30-m string

D. none: no right angle

Answer it.

Whiteboards · Question 6 of 9

A kite flies on 30 m of string, straight above a spot 18 m from where the string is held.

Which length is the hypotenuse?

The square corner is under the kite.

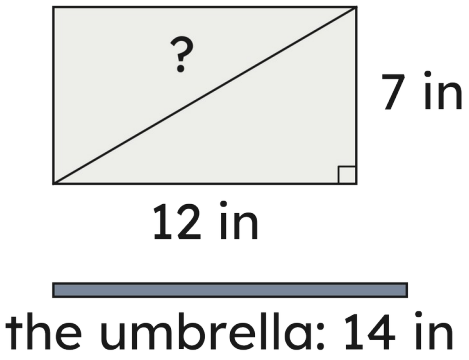
Across from it: the string.

Answer: C — the 30-m string

Whiteboards · Question 7 of 9

A box is 7 inches wide and 12 inches long.

Will a 14-in umbrella lie flat in it, corner to corner?

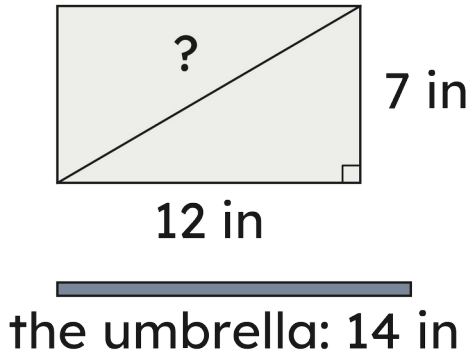


Answer it.

Whiteboards · Question 7 of 9

A box is 7 inches wide and 12 inches long.

Will a 14-in umbrella lie flat in it, corner to corner?



$$c^2 = 49 + 144 = 193$$

$$c = \sqrt{193} \approx 13.9$$

Answer: No — the diagonal is about 13.9 in.

Whiteboards · Question 8 of 9

A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall.

How high does it reach, to the nearest tenth of a foot?

Answer it.

Whiteboards · Question 8 of 9

A 10-ft ladder that weighs 25 pounds has its foot 7 ft from a wall.

How high does it reach, to the nearest tenth of a foot?

$$h^2 = 100 - 49 = 51$$

$$h = \sqrt{51} \approx 7.1$$

Answer: About 7.1 ft

Whiteboards · Question 9 of 9

A window is 12 ft above the ground.

Why can an 11-ft ladder never reach it, wherever its foot stands?

Write your answer in sentences.

Whiteboards · Question 9 of 9

A window is 12 ft above the ground.

Why can an 11-ft ladder never reach it, wherever its foot stands?

The ladder is the hypotenuse, the longest side.

The height is a leg, so it is less than 11 ft.

Answer: The height is a leg — always shorter than the 11-ft ladder.

Independent Set

1. A tablet screen is 5 inches wide and 8 inches tall. How long is its diagonal, to the nearest tenth of an inch?

2. A fire truck's 50-ft ladder reaches 48 ft up a building. How far from the building is its foot?

3. A ramp rises 3 ft over a run of 14 ft. How long is the ramp, to the nearest tenth of a foot?

4. A park is 7 m by 11 m. How much shorter is the walk straight across, corner to corner, than the walk along two sides? Give it to the nearest tenth of a meter.

5. Will a 10-in pencil lie flat in a box 7 inches by 8 inches, corner to corner?

6. A kite flies at the end of 25 m of string, straight above a spot 20 m from where the string is held. How high is the kite?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Pythagorean theorem: word problems — 87U

GRADE 7 ACCELERATED · UNIT 6 · LESSON 5

Distance on the Coordinate Plane

MA.8.GR.1.2

I can find the distance between two points: subtract on a line, and draw the right triangle when the segment is slanted.

Warm-Up

1. A 25-ft ladder reaches 20 ft up a wall. How far out is its foot?
2. Do sides of 12, 35 and 37 make a right triangle?
3. A map's scale is 1 cm : 5 km. Towns 3.4 cm apart are how far?
4. How far apart are $(2, 5)$ and $(2, -3)$?

Warm-Up

1. A 25-ft ladder reaches 20 ft up a wall. How far out is its foot?

   15 ft

2. Do sides of 12, 35 and 37 make a right triangle?

   Yes — $144 + 1{,}225 = 1{,}369 = 37^2$

3. A map's scale is 1 cm : 5 km. Towns 3.4 cm apart are how far?

   17 km

4. How far apart are $(2, 5)$ and $(2, -3)$? **8**

Vocabulary

horizontal distance

between two points with the same y: the difference of the x's

vertical distance

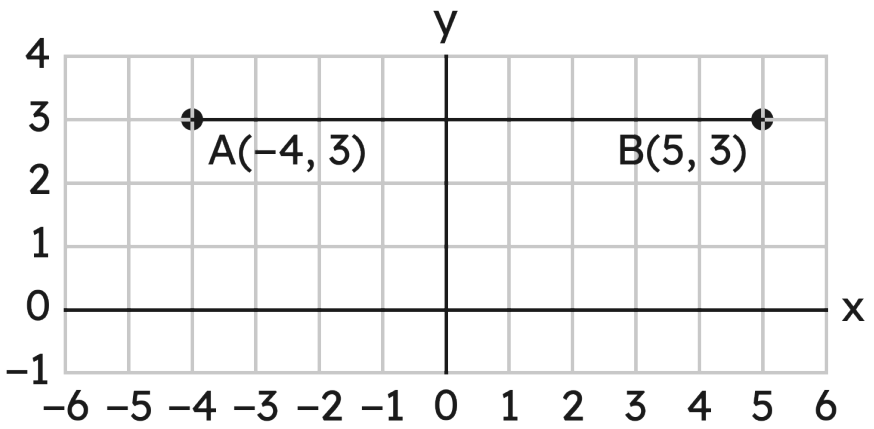
between two points with the same x: the difference of the y's

legs on the grid

the horizontal and vertical segments under a slanted one; the slanted segment is the hypotenuse

Notes

I. On one line: subtract — and across zero, add

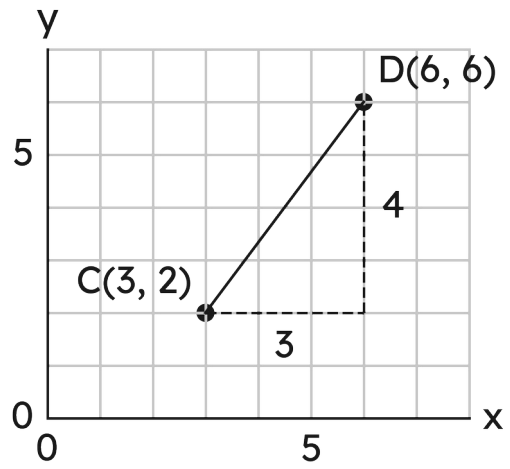


Same y: subtract the x's. Same x: subtract the y's.

$5 - (-4) = 9$: across zero, 4 and 5 add.

Notes

II. Slanted: draw the right triangle

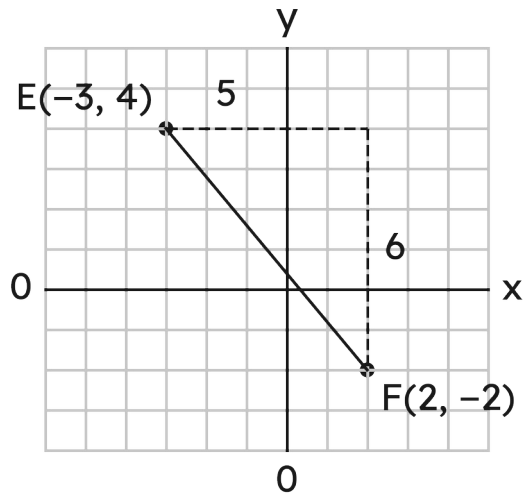


$$c^2 = 3^2 + 4^2 = 25, \quad c = 5$$

The legs are the across and the up. The distance is the hypotenuse.

Notes

III. The legs come from the coordinates



across $2-(-3)=5$ down $4-(-2)=6$

$$c^2 = 25 + 36 = 61,\ c = \sqrt{61} \approx 7.8$$

Example 1

Find the distance between $(-6, -2)$ and $(-6, 7)$.

Example 1

$$7 - (-2) = 9$$

$$2 + 7 = 9$$

Answer: 9 units

Your Turn

Find the distance between $(-3, 5)$ and $(8, 5)$.

Your Turn

Find the distance between $(-3, 5)$ and $(8, 5)$.

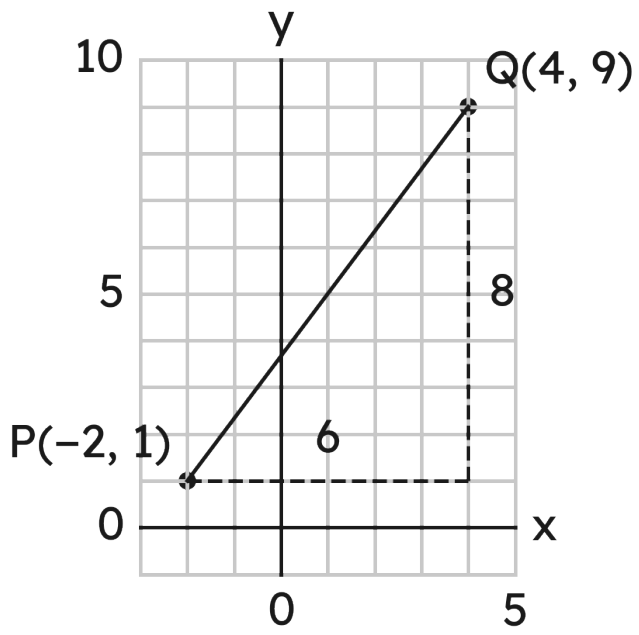
$$8 - (-3) = 11$$

check: $3 + 8 = 11$

Answer: 11 units

Example 2

Find the distance between $(-2, 1)$ and $(4, 9)$.



Example 2

$4 - (-2) = 6, \quad 9 - 1 = 8$

$c^2 = 36 + 64 = 100$

$c = \sqrt{100} = 10$

Answer: 10 units

Your Turn

Find the distance between $(-5, -3)$ and $(2, 1)$, to the nearest tenth.

Your Turn

Find the distance between $(-5, -3)$ and $(2, 1)$, to the nearest tenth.

$$\text{across} \quad 2-(-5)=7, \quad \text{up} \quad 1-(-3)=4$$

$$c^2 = 49 + 16 = 65, \quad c = \sqrt{65} \approx 8.1$$

Answer: $\sqrt{65} \approx 8.1$ units

Whiteboards · Question 1 of 9

How far apart are $(-7, -4)$ and $(-7, 6)$?

Answer it.

Whiteboards · Question 1 of 9

How far apart are $(-7, -4)$ **and** $(-7, 6)$**?**

$$6 - (-4) = 10$$

Answer: 10 units

Whiteboards · Question 2 of 9

How far apart are $(-9, 2)$ and $(3, 2)$?

Answer it.

Whiteboards · Question 2 of 9

How far apart are $(-9, 2)$ **and** $(3, 2)$**?**

$$3 - (-9) = 12$$

Answer: 12 units

Whiteboards · Question 3 of 9

What is the distance between $(-5, 1)$ and $(3, 1)$?

A. 9

B. 8

C. 2

D. −8

Answer it.

Whiteboards · Question 3 of 9

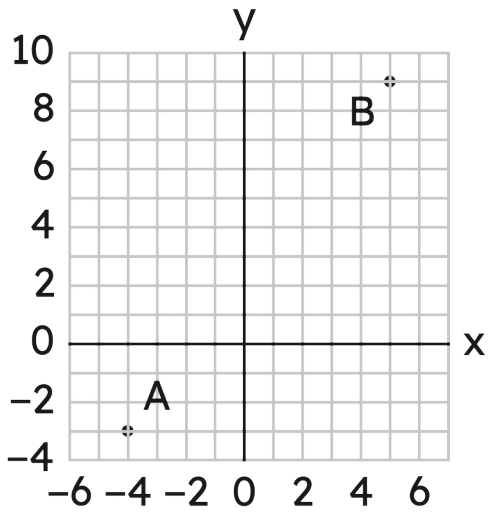
What is the distance between $(-5, 1)$ **and** $(3, 1)$**?**

$$3 - (-5) = 8$$

Answer: B — 8 units

Whiteboards · Question 4 of 9

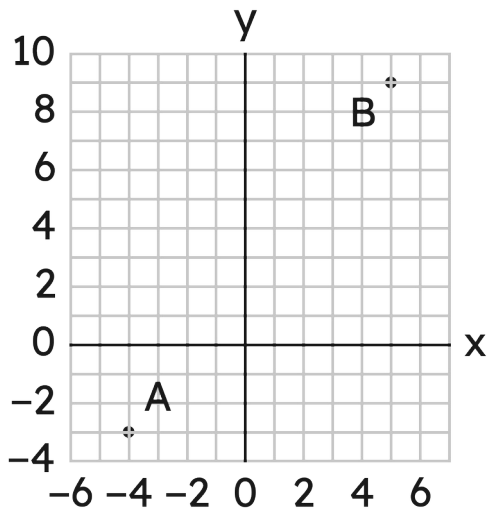
How far apart are $A(-4, -3)$ **and** $B(5, 9)$**?**



Answer it.

Whiteboards · Question 4 of 9

How far apart are $A(-4, -3)$ **and** $B(5, 9)$**?**



across 9, up 12

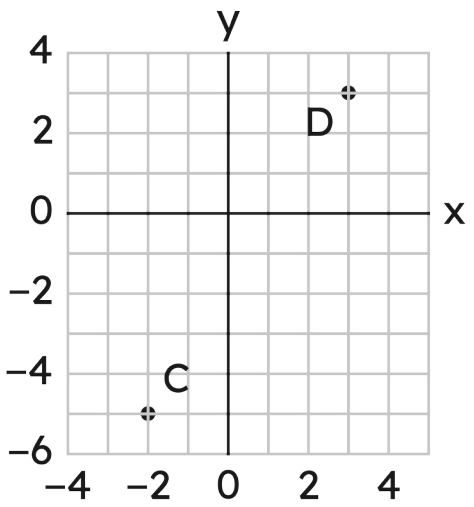
$c^2 = 81 + 144 = 225$

$c = 15$

Answer: 15 units

Whiteboards · Question 5 of 9

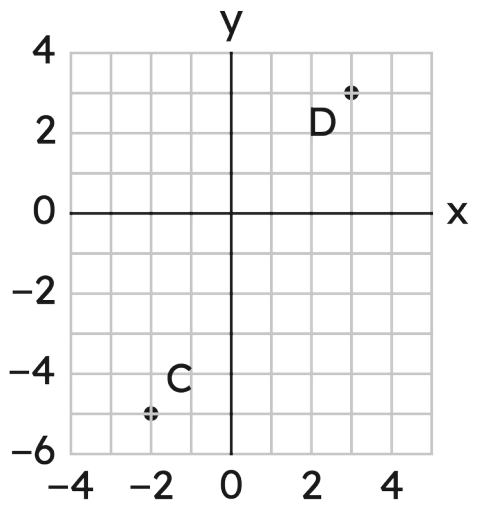
How far apart are $C(-2, -5)$ and $D(3, 3)$, to the nearest tenth?



Answer it.

Whiteboards · Question 5 of 9

How far apart are $C(-2, -5)$ **and** $D(3, 3)$**, to the nearest tenth?**



across 5, up 8

$c^2 = 25 + 64 = 89$

$c = \sqrt{89} \approx 9.4$

Answer: $\sqrt{89} \approx 9.4$ units

Whiteboards · Question 6 of 9

Which gives the distance between $(4, 5)$ and $(-3, -1)$?

A. $\sqrt{1^2 + 6^2}$

B. $\sqrt{4^2 + 5^2}$

C. $\sqrt{7^2 + 6^2}$

D. $7 + 6$

Answer it.

Whiteboards · Question 6 of 9

Which gives the distance between $(4, 5)$ and $(-3, -1)$?

across $4 - (-3) = 7$, up $5 - (-1) = 6$

$$\sqrt{7^2 + 6^2} = \sqrt{85}$$

Answer: C — $\sqrt{(7^2 + 6^2)} = \sqrt{85}$

Whiteboards · Question 7 of 9

How far is $(5, -12)$ from the origin?

Answer it.

Whiteboards · Question 7 of 9

How far is $(5, -12)$ from the origin?

across 5, down 12

$c^2 = 25 + 144 = 169$, $c = 13$

Answer: 13 units

Whiteboards · Question 8 of 9

On a town map in miles, the school is at $(-3, 2)$, the library at $(4, 2)$ and the town hall at $(0, 5)$.

How far is the school from the library?

Answer it.

Whiteboards · Question 8 of 9

On a town map in miles, the school is at $(-3, 2)$, the library at $(4, 2)$ and the town hall at $(0, 5)$.

How far is the school from the library?

$$4 - (-3) = 7$$

Answer: 7 mi

Whiteboards · Question 9 of 9

Ty says $(-3, 0)$ and $(7, 0)$ are 4 units apart.

What did he do? What is the distance?

Write your answer in sentences.

Whiteboards · Question 9 of 9

Ty says $(-3, 0)$ and $(7, 0)$ are 4 units apart.

What did he do? What is the distance?

$$7 - (-3) = 10$$

check: $3 + 7 = 10$

Answer: He subtracted 7 – 3. Across zero the pieces add: 10 units.

Independent Set

1. Find the distance between $(-6, 3)$ and $(7, 3)$.
2. Find the distance between $(5, -7)$ and $(5, 4)$.
3. Find the distance between $(-7, 1)$ and $(5, -4)$.
4. Find the distance between $(-2, 5)$ and $(1, -2)$, to the nearest tenth.
5. How far is $(-4, 9)$ from the origin, to the nearest tenth?
6. On a map in kilometers, a pool is at $(-2, -3)$ and a park at $(7, 9)$. How far apart are they in a straight line?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Find the distance between two points on a horizontal or vertical line — L9V

Find the distance between two points — ZBP

GRADE 7 ACCELERATED · UNIT 6 · LESSON 6

Real-World Coordinate-Plane Problems; Perimeter

MA.8.GR.1.2

I can find the perimeter of a figure on the coordinate plane, and a straight-line distance on a map.

Warm-Up

1. How far apart are $(-2, -3)$ and $(4, 5)$?
2. Legs of 2 m and 7 m. The hypotenuse, to the nearest tenth?
3. A map's scale is 1 in : 30 mi. Towns 2.5 in apart are how far?
4. The rectangle $(-3, 2), (4, 2), (4, -1), (-3, -1)$: its perimeter?

Warm-Up

1. How far apart are $(-2, -3)$ and $(4, 5)$? **10 units**
2. Legs of 2 m and 7 m. The hypotenuse, to the nearest tenth?

   $\sqrt{53} \approx 7.3$ m
3. A map's scale is 1 in : 30 mi. Towns 2.5 in apart are how far?

   75 mi
4. The rectangle $(-3, 2), (4, 2), (4, -1), (-3, -1)$: its perimeter?

   20 units

Vocabulary

perimeter

the distance around a figure: the sum of all its sides

height of a triangle

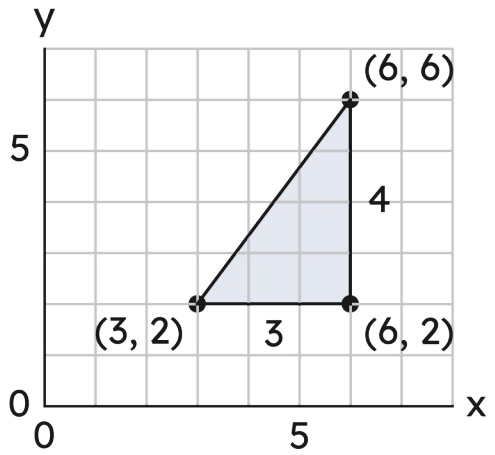
the segment from a corner straight to the opposite side, meeting it at a right angle

straight-line distance

the length of the one segment between two points — on a map, straight across, not along the streets

Notes

I. Perimeter: every side — a slanted one by its triangle

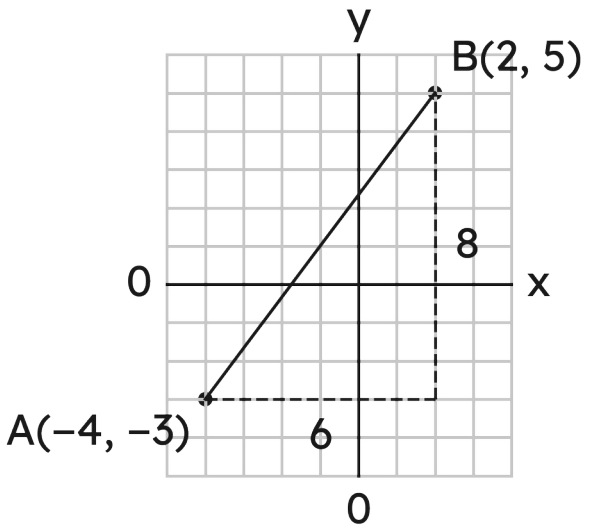


slanted side: $c^2 = 3^2 + 4^2 = 25, \quad c = 5$

$$P = 3 + 4 + 5 = 12$$

Notes

II. Along the streets, or straight

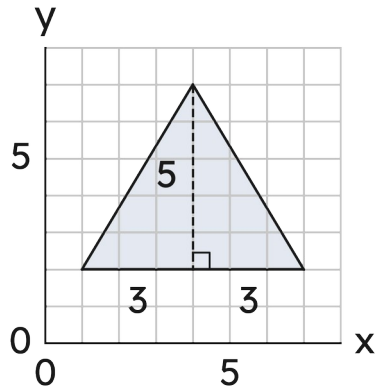


streets: $6 + 8 = 14$ blocks

straight: $c^2 = 36 + 64 = 100$, $c = 10$ blocks

Notes

III. No right angle: split it — and round once, at the end



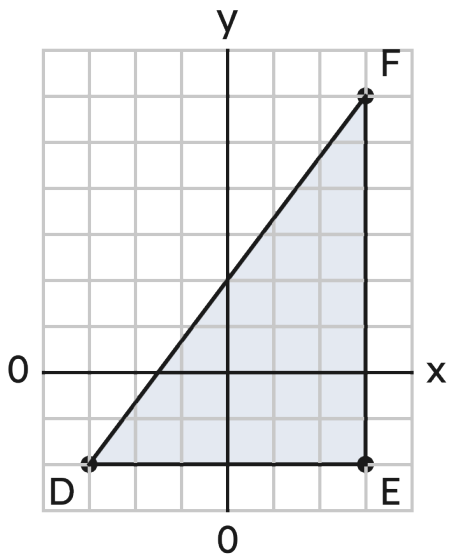
each slanted side: $c^2 = 3^2 + 5^2 = 34$, $c = \sqrt{34} \approx 5.831$

$$P \approx 6 + 5.831 + 5.831 = 17.662 \approx 17.7$$

Round once, at the end: 6 + 5.8 + 5.8 = 17.6 is short.

Example 1

Triangle DEF has corners $D(-3, -2)$, $E(3, -2)$ and $F(3, 6)$.



Find its perimeter.

Example 1

$3 - (-3) = 6, \quad 6 - (-2) = 8$

$c^2 = 36 + 64 = 100, \quad c = 10$

$P = 6 + 8 + 10 = 24$

Answer: 24 units

Your Turn

Find the perimeter of the triangle with corners

$(-5, -7)$, $(4, -7)$ and $(4, 5)$.

Your Turn

Find the perimeter of the triangle with corners

$(-5, -7)$, $(4, -7)$ and $(4, 5)$.

across $4 - (-5) = 9$, up $5 - (-7) = 12$

$c^2 = 81 + 144 = 225$, $c = 15$

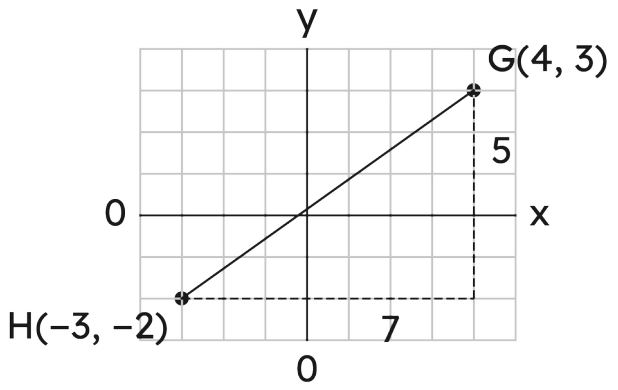
$P = 9 + 12 + 15 = 36$

Answer: 36 units

Example 2

On a town map in blocks, Ana's home H is at $(-3, -2)$

and the gym G at $(4, 3)$.



How far along the streets? In a straight line, to the nearest tenth?

Example 2

$4 - (-3) = 7, \quad 3 - (-2) = 5$

$7 + 5 = 12$

$c^2 = 49 + 25 = 74, \quad c = \sqrt{74} \approx 8.6$

Answer: 12 blocks along the streets; about 8.6 blocks straight

Your Turn

On a town map in blocks, Ben's home is at $(-5, 4)$ and the park at $(-2, -4)$. How far is it along the streets?

In a straight line, to the nearest tenth?

Your Turn

On a town map in blocks, Ben's home is at $(-5, 4)$

and the park at $(-2, -4)$. How far is it along the streets?

In a straight line, to the nearest tenth?

across 3, down 8: streets $3 + 8 = 11$

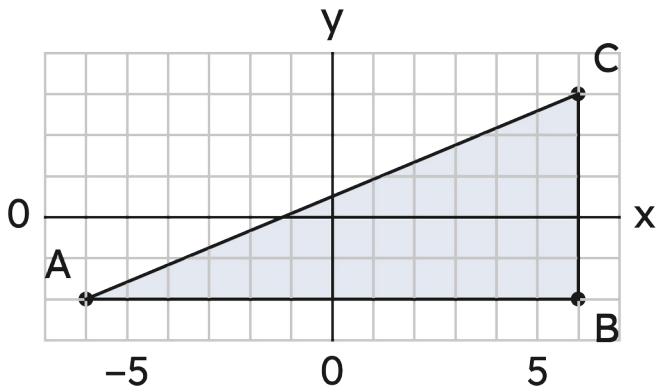
straight: $c^2 = 9 + 64 = 73$, $c = \sqrt{73} \approx 8.5$

Answer: 11 blocks along the streets; about 8.5 blocks straight

Whiteboards · Question 1 of 9

Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$.

Find its perimeter.

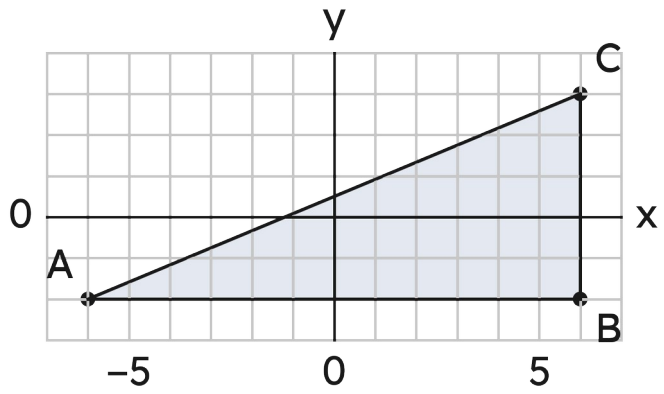


Answer it.

Whiteboards · Question 1 of 9

Triangle ABC has corners $A(-6, -2)$, $B(6, -2)$ and $C(6, 3)$.

Find its perimeter.



$6 - (-6) = 12$, $3 - (-2) = 5$

$c^2 = 144 + 25 = 169$, $c = 13$

$P = 12 + 5 + 13 = 30$

Answer: 30 units

Whiteboards · Question 2 of 9

A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$.

Find its perimeter, to the nearest tenth.

Answer it.

Whiteboards · Question 2 of 9

A triangle has corners $(-3, -4)$, $(1, -4)$ and $(1, 5)$.

Find its perimeter, to the nearest tenth.

$$1 - (-3) = 4, \quad 5 - (-4) = 9$$

$$c^2 = 16 + 81 = 97, \quad c = \sqrt{97}$$

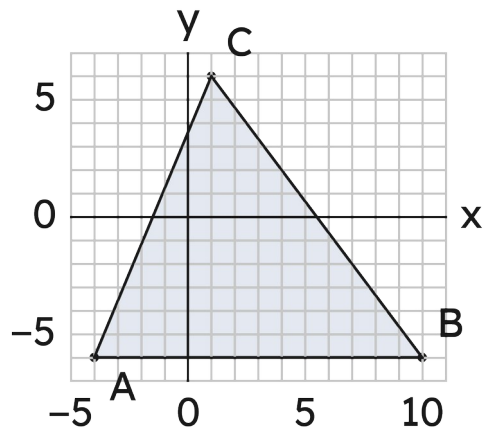
$$P = 4 + 9 + \sqrt{97} \approx 22.8$$

Answer: 22.8 units

Whiteboards · Question 3 of 9

Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle.

Find its perimeter.

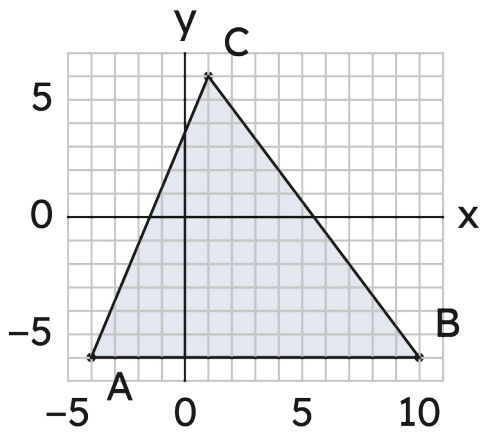


Answer it.

Whiteboards · Question 3 of 9

Triangle $A(-4, -6)$, $B(10, -6)$, $C(1, 6)$ has no right angle.

Find its perimeter.



AC: $c^2 = 5^2 + 12^2 = 169$, $c = 13$

BC: $c^2 = 9^2 + 12^2 = 225$, $c = 15$

$P = 14 + 13 + 15 = 42$

Answer: 42 units

Whiteboards · Question 4 of 9

What is the perimeter of the triangle with corners $(-2, 3)$, $(2, 3)$ and $(2, -4)$, to the nearest tenth?

A. 11

B. 22

C. 76

D. 19.1

Answer it.

Whiteboards · Question 4 of 9

What is the perimeter of the triangle with corners $(-2, 3)$, $(2, 3)$ and $(2, -4)$, to the nearest tenth?

4 and 7 on grid lines

$c^2 = 16 + 49 = 65, \quad c = \sqrt{65}$

$P = 4 + 7 + \sqrt{65} \approx 19.1$

Answer: D — 19.1 units

Whiteboards · Question 5 of 9

On a map in blocks, the library is at $(-7, -1)$

and the pool at $(5, 4)$.

How much shorter is the straight line than the streets?

Answer it.

Whiteboards · Question 5 of 9

On a map in blocks, the library is at $(-7, -1)$

and the pool at $(5, 4)$.

How much shorter is the straight line than the streets?

streets: $12 + 5 = 17$

straight: $c^2 = 144 + 25 = 169$, $c = 13$

$17 - 13 = 4$

Answer: 4 blocks shorter

Whiteboards · Question 6 of 9

How long is a diagonal of the rectangle with corners $(-3, -4)$, $(6, -4)$, $(6, 8)$ and $(-3, 8)$?

A. 15

B. 21

C. 42

D. 10

Answer it.

Whiteboards · Question 6 of 9

How long is a diagonal of the rectangle with corners

$(-3, -4)$, $(6, -4)$, $(6, 8)$ **and** $(-3, 8)$**?**

sides $6 - (-3) = 9$ and $8 - (-4) = 12$

$c^2 = 81 + 144 = 225$, $c = 15$

Answer: A — 15 units

Whiteboards · Question 7 of 9

On a park map in meters, the fountain is at $(-5, 2)$

and the gate at $(4, -3)$.

How far is it straight across, to the nearest tenth?

Answer it.

Whiteboards · Question 7 of 9

On a park map in meters, the fountain is at $(-5, 2)$ and the gate at $(4, -3)$.

How far is it straight across, to the nearest tenth?

across $4 - (-5) = 9$, down $2 - (-3) = 5$

$c^2 = 81 + 25 = 106$, $c = \sqrt{106} \approx 10.3$

Answer: $\sqrt{106} \approx 10.3$ m

Whiteboards · Question 8 of 9

On a trail map in miles, the camp is at $(-3, -6)$,

the lake at $(2, 6)$ and the trailhead at $(-1, 0)$.

How far is the camp from the lake, in a straight line?

Answer it.

Whiteboards · Question 8 of 9

On a trail map in miles, the camp is at $(-3, -6)$,

the lake at $(2, 6)$ and the trailhead at $(-1, 0)$.

How far is the camp from the lake, in a straight line?

across $2 - (-3) = 5$, up $6 - (-6) = 12$

$c^2 = 25 + 144 = 169$, $c = 13$

Answer: 13 mi

Whiteboards · Question 9 of 9

Mia says the segment from $(0, 0)$ to $(3, 1)$ is 3 units long,

because it crosses 3 grid squares.

Is she right? Explain, and find its length to the nearest tenth.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Mia says the segment from $(0, 0)$ to $(3, 1)$ is 3 units long,

because it crosses 3 grid squares.

Is she right? Explain, and find its length to the nearest tenth.

across 3, up 1

$c^2 = 9 + 1 = 10, \quad c = \sqrt{10} \approx 3.2$

Answer: No: it is the hypotenuse of a triangle with legs 3 and 1, √10 ≈ 3.2 units.

Independent Set

1. Find the perimeter of the triangle $(-2, -7)$, $(-2, 5)$, $(3, 5)$.
2. Find the perimeter of the triangle $(-4, 1)$, $(2, 1)$, $(2, -4)$, to the nearest tenth.
3. Find the perimeter of the parallelogram $(-4, -5)$, $(2, -5)$, $(7, 7)$, $(1, 7)$.
4. On a map in blocks, Kai's home is at $(-6, -5)$ and his school at $(3, 7)$. How much shorter is the straight line than the streets?
5. On a map in miles, a dock is at $(-4, 3)$ and an island at $(3, -5)$. How far apart are they, to the nearest tenth?
6. A rectangle has corners $(-3, -2)$, $(2, -2)$, $(2, 10)$ and $(-3, 10)$. How long is a diagonal?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Pythagorean theorem: find the perimeter — VGE

GRADE 7 ACCELERATED · UNIT 6 · LESSONS 7–8

Similar Triangles; Indirect Measurement

MA.8.GR.2.4

I can find a missing side of similar triangles with the scale factor, and use similar triangles to measure what I cannot reach.

Warm-Up

1. The triangle $(0, 0)$, $(3, 0)$, $(3, 2)$: its perimeter, to the tenth?
2. A 13-ft ladder's foot is 5 ft from a wall. How high does it reach?
3. A 10 cm side is 4 cm in a copy. What is the scale factor?
4. At 1 in : 6 ft, a 3.5-in wall on a drawing is how long?

Warm-Up

1. The triangle $(0, 0)$, $(3, 0)$, $(3, 2)$: its perimeter, to the tenth?

   8.6 units

2. A 13-ft ladder's foot is 5 ft from a wall. How high does it reach?

   12 ft

3. A 10 cm side is 4 cm in a copy. What is the scale factor?

   k = 0.4

4. At 1 in : 6 ft, a 3.5-in wall on a drawing is how long? **21 ft**

Vocabulary

similar triangles

the same shape, any size: matching angles equal, and every side multiplied by the same number

corresponding sides

sides that match — across from equal angles: shortest with shortest, longest with longest

scale factor k

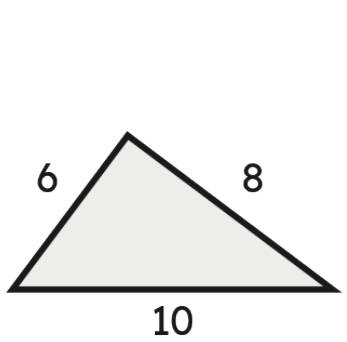
copy ÷ original, for one matching pair of sides

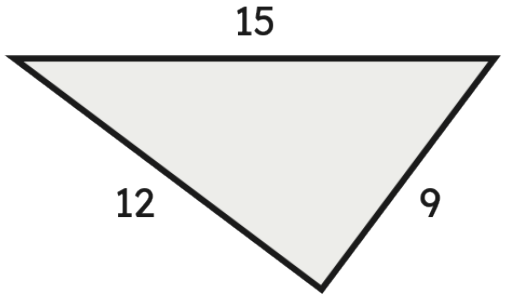
indirect measurement

finding a length you cannot reach from similar triangles you can measure

Notes

I. Similar: every side times the same k



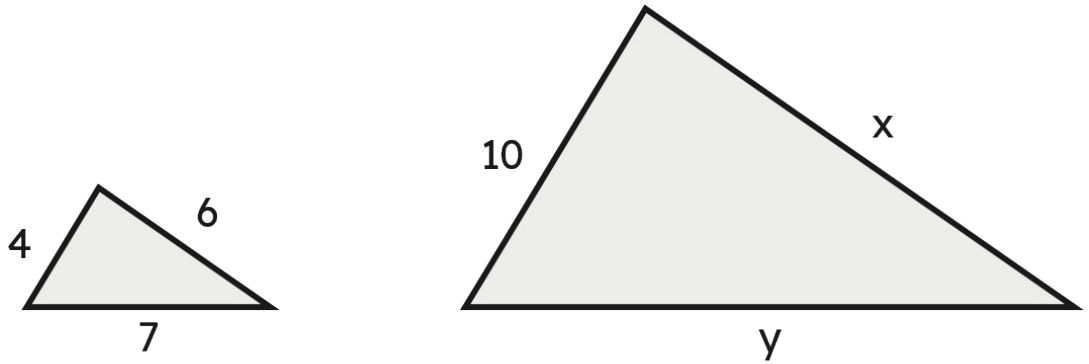


$$\frac{9}{6} = 1.5, \quad \frac{12}{8} = 1.5, \quad \frac{15}{10} = 1.5$$

Three ratios, all equal: similar. Match shortest to shortest, longest to longest.

Notes

II. A missing side: find k, then multiply

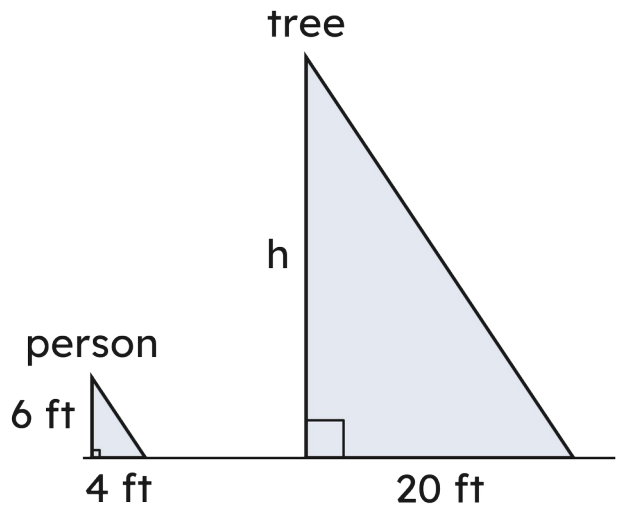


$$k = \frac{10}{4} = 2.5$$

$$x = 6 \times 2.5 = 15, \quad y = 7 \times 2.5 = 17.5$$

Notes

III. Same sun, same angle: shadows

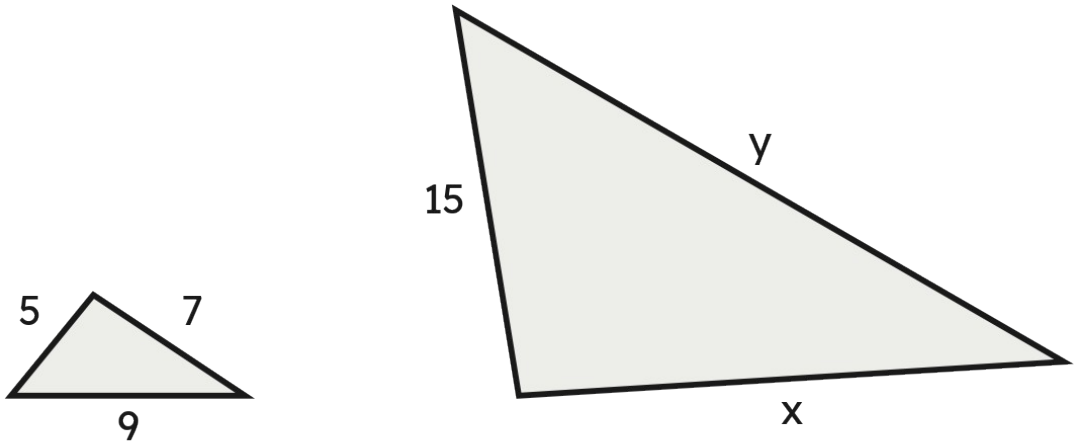


$$k = \frac{20}{4} = 5$$

$$h = 6 \times 5 = 30 \text{ ft}$$

Example 1

The two triangles are similar.



Find x and y.

Example 1

$$k = \frac{15}{5} = 3$$

$$x = 7 \times 3 = 21$$

$$y = 9 \times 3 = 27$$

Answer: x = 21, y = 27

Your Turn

A triangle has sides 3, 5 and 7.

A similar triangle has sides 28, x and 12. Find x.

Your Turn

A triangle has sides 3, 5 and 7.

A similar triangle has sides 28, x and 12. Find x.

sorted: 3, 5, 7 and 12, x, 28

$$k = \frac{12}{3} = 4$$

$$x = 5 \times 4 = 20$$

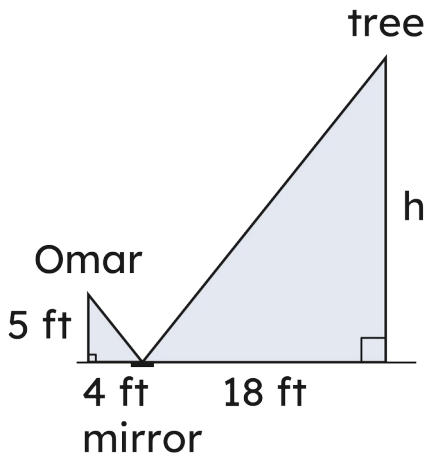
Answer: x = 20

Example 2

Omar's eyes are 5 ft up, 4 ft from a mirror on the ground.

In it he sees the top of a tree 18 ft beyond the mirror.

The two right triangles are similar.



How tall is the tree?

Example 2

$$k = \frac{18}{4} = 4.5$$

$$h = 5 \times 4.5 = 22.5$$

Answer: The tree is 22.5 ft tall.

Your Turn

Ana's eyes are 4 ft up, 1.5 ft from a mirror on the ground.

In it she sees the top of a flagpole 12 ft beyond the mirror.

The two right triangles are similar. How tall is the flagpole?

Your Turn

Ana's eyes are 4 ft up, 1.5 ft from a mirror on the ground.

In it she sees the top of a flagpole 12 ft beyond the mirror.

The two right triangles are similar. How tall is the flagpole?

$$k = \frac{12}{1.5} = 8$$

$$h = 4 \times 8 = 32$$

Answer: 32 ft

Whiteboards · Question 1 of 9

Are triangles with sides 8, 12, 14 and 12, 18, 21 similar?

Answer it.

Whiteboards · Question 1 of 9

Are triangles with sides 8, 12, 14 and 12, 18, 21 similar?

$$\frac{12}{8} = 1.5, \quad \frac{18}{12} = 1.5, \quad \frac{21}{14} = 1.5$$

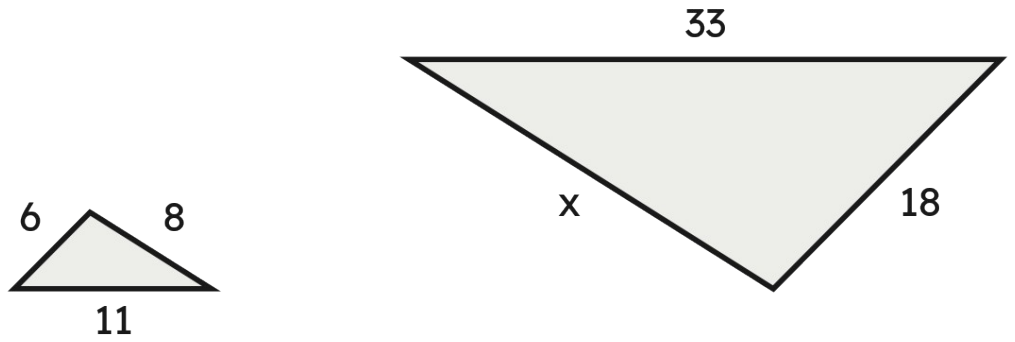
all three equal

Answer: Yes — every ratio is 1.5

Whiteboards · Question 2 of 9

The two triangles are similar.

Find x.

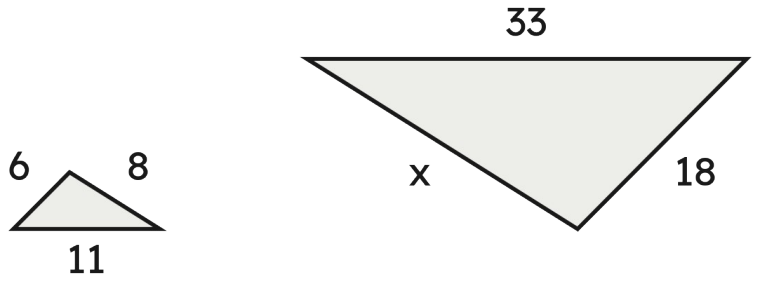


Answer it.

Whiteboards · Question 2 of 9

The two triangles are similar.

Find x.



sorted: 6, 8, 11 and 18, x, 33

$$k = \frac{18}{6} = 3$$

$$x = 8 \times 3 = 24$$

Answer: x = 24

Whiteboards · Question 3 of 9

A triangle has sides of 12, 16 and 20 inches.

A similar triangle's longest side is 5 inches.

Find its other two sides.

Answer it.

Whiteboards · Question 3 of 9

A triangle has sides of 12, 16 and 20 inches.

A similar triangle's longest side is 5 inches.

Find its other two sides.

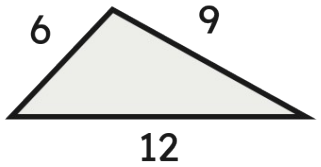
$$k = \frac{5}{20} = 0.25$$

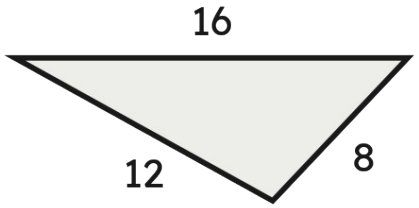
$$12 \times 0.25 = 3, \quad 16 \times 0.25 = 4$$

Answer: 3 inches and 4 inches

Whiteboards · Question 4 of 9

Similar triangles: which side of the second matches the 9?





A. 8

B. 16

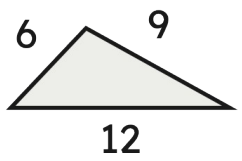
C. 12

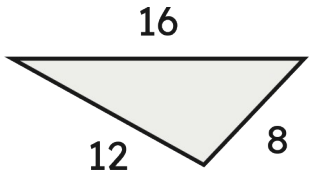
D. none of them

Answer it.

Whiteboards · Question 4 of 9

Similar triangles: which side of the second matches the 9?





sorted: 6, 9, 12 and 8, 12, 16

the middle sides match: 9 and 12

Answer: C — the 12

Whiteboards · Question 5 of 9

Which triangle is similar to one with sides 5, 6 and 8?

A. 7, 8, 10

B. 10, 12, 16

C. 10, 12, 15

D. 5, 6, 9

Answer it.

Whiteboards · Question 5 of 9

Which triangle is similar to one with sides 5, 6 and 8?

$$\frac{10}{5} = 2, \quad \frac{12}{6} = 2, \quad \frac{16}{8} = 2$$

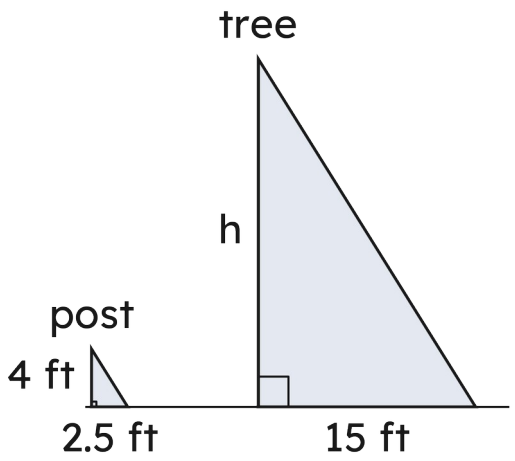
one k for every side

Answer: B — 10, 12, 16

Whiteboards · Question 6 of 9

A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow.

How tall is the tree?

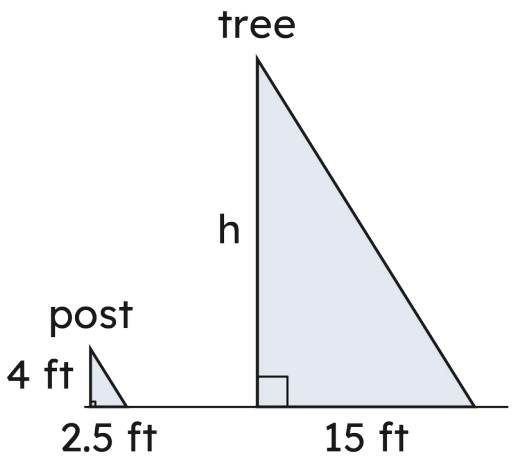


Answer it.

Whiteboards · Question 6 of 9

A 4-ft post casts a 2.5-ft shadow as a tree casts a 15-ft shadow.

How tall is the tree?



$$k = \frac{15}{2.5} = 6$$

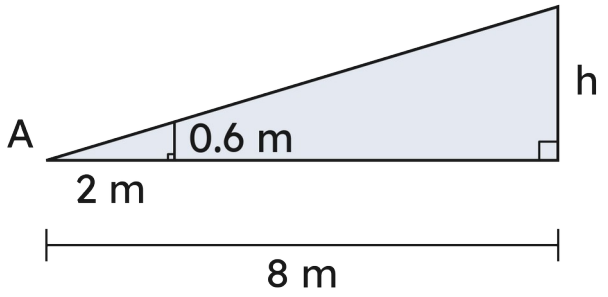
$$h = 4 \times 6 = 24$$

Answer: 24 ft

Whiteboards · Question 7 of 9

A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar.

How high is the ramp 8 m from A?

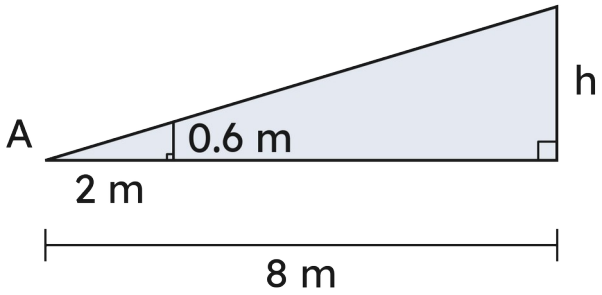


Answer it.

Whiteboards · Question 7 of 9

A ramp rises 0.6 m over its first 2 m from A. The right triangles are similar.

How high is the ramp 8 m from A?



$$k = \frac{8}{2} = 4$$

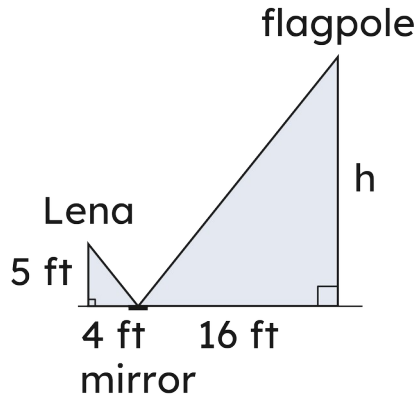
$$h = 0.6 \times 4 = 2.4$$

Answer: 2.4 m

Whiteboards · Question 8 of 9

Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror.

The two right triangles are similar. How tall is the flagpole?

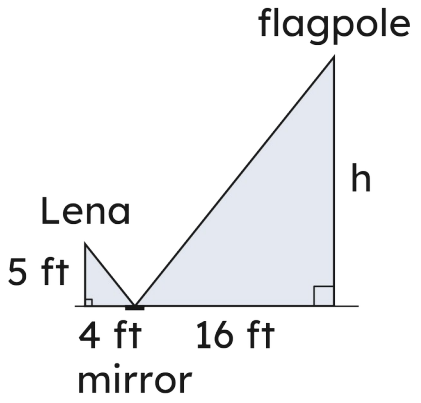


Answer it.

Whiteboards · Question 8 of 9

Lena's eyes are 5 ft up. In a mirror on the ground 4 ft away she sees the top of a 45-year-old flagpole, 16 ft past the mirror.

The two right triangles are similar. How tall is the flagpole?



$$k = \frac{16}{4} = 4$$

$$h = 5 \times 4 = 20$$

Answer: 20 ft

Whiteboards · Question 9 of 9

Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2.

Is she right? Explain.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Kim says triangles with sides 3, 4, 5 and 5, 6, 7 are similar, because each side grew by 2.

Is she right? Explain.

$\frac{5}{3} \approx 1.67$, $\frac{6}{4} = 1.5$, $\frac{7}{5} = 1.4$: three ratios

Answer: No — similar means × one k, not + 2.

Independent Set

1. Are triangles with sides 4, 10, 12 and 6, 15, 20 similar? Show the three ratios.

2. A triangle has sides 8, 10 and 13. A similar triangle's 10-side is 25. Find its other two sides.

3. A triangle has sides 15, 25 and 30. A similar triangle's longest side is 6. Find its other two sides.

4. A 1.5-m child casts a 2-m shadow. At the same moment a tree casts a 14-m shadow. How tall is the tree?

5. The two right triangles are similar and share corner A. The small one runs 3 m along the ground from A and stands 1.2 m high. The big one runs 12 m along the ground from A. How high is it?

6. Which is similar to a triangle with sides 8, 10 and 14: one with sides 10, 12, 16, or one with sides 12, 15, 21? Show why.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Side lengths of similar triangles — QSK

Similar triangles and indirect measurement — 88W

Identify similar triangles — 6PD

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 1

Angle Pairs: Supplementary, Complementary, Vertical, Adjacent

MA.8.GR.1.4

I can name an angle pair and use it to find a missing angle, with an equation when the angles are expressions.

Warm-Up

1. A 6-ft man casts a 4-ft shadow; a tree, 18 ft. The tree's height?
2. How far apart are $(-4, 2)$ and $(5, 2)$?
3. Ana types 210 words in 5 minutes. Words per minute?
4. Two angles make a straight angle. One is 130°. The other?

Warm-Up

1. A 6-ft man casts a 4-ft shadow; a tree, 18 ft. The tree's height?

   27 ft

2. How far apart are $(-4, 2)$ and $(5, 2)$? **9 units**

3. Ana types 210 words in 5 minutes. Words per minute? **42**

4. Two angles make a straight angle. One is 130°. The other?

   50°

Vocabulary

adjacent angles

two angles with the same vertex and a shared side, that do not overlap

vertical angles

the angles across from each other where two lines cross; they are equal

complementary angles

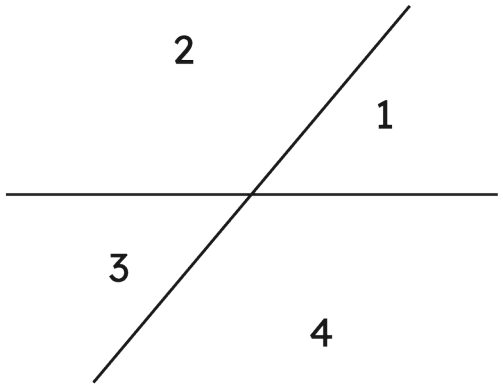
two angles that add to 90°

supplementary angles

two angles that add to 180°; a linear pair (adjacent, on a straight line) is supplementary

Notes

I. Adjacent and vertical

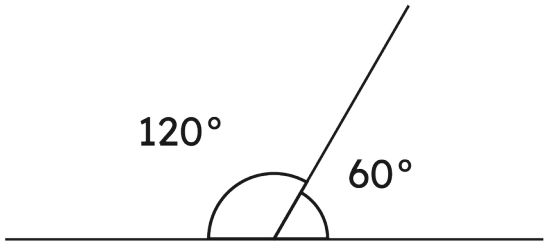


Adjacent: one vertex, a shared side, no overlap — angles 1 and 2.

Vertical: across, and equal — $m\angle 1 = m\angle 3$ **(m for measure).**

Notes

II. Complementary and supplementary

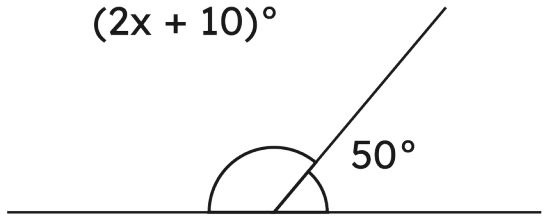


Complementary: add to 90°. Supplementary: add to 180°.

A linear pair is supplementary: 60° + 120° = 180°.

Notes

III. Name the pair, write the equation, put x back

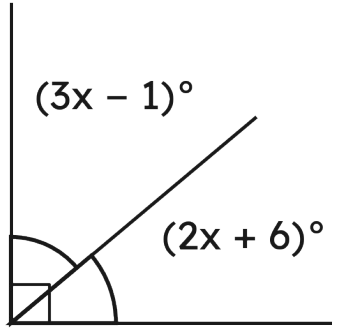


$$2x + 10 + 50 = 180, \quad 2x = 120, \quad x = 60$$

the angle: $2(60) + 10 = 130$

Example 1

A ray splits a right angle into $(3x - 1)°$ and $(2x + 6)°$.



Find x and both angles.

Example 1

$3x - 1 + 2x + 6 = 90$

$5x + 5 = 90, \quad x = 17$

$3(17) - 1 = 50, \quad 2(17) + 6 = 40$

Answer: x = 17; the angles are 50° and 40°

Your Turn

$(4x)°$ and $(x + 10)°$ are complementary.

Find x and both angles.

Your Turn

$(4x)°$ and $(x + 10)°$ are complementary.

Find x and both angles.

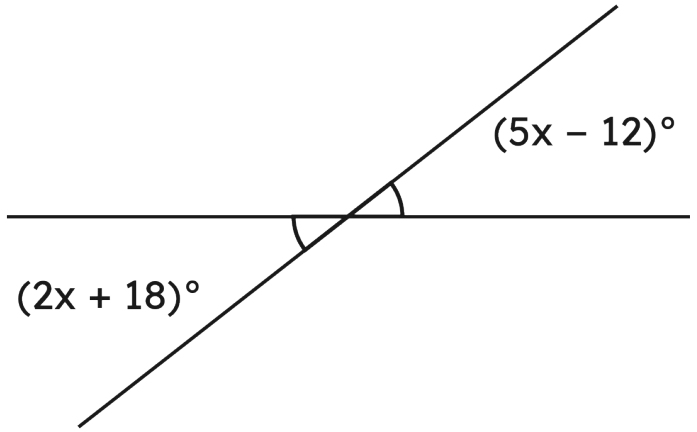
$$4x + x + 10 = 90, \quad 5x = 80, \quad x = 16$$

$$4(16) = 64, \quad 16 + 10 = 26$$

Answer: x = 16; 64° and 26°

Example 2

Two lines cross. Vertical angles measure $(5x - 12)°$ and $(2x + 18)°$.



Find the measure of each angle.

Example 2

$5x - 12 = 2x + 18$

$3x - 12 = 18$

$3x = 30, \quad x = 10$

$5(10) - 12 = 38$

Answer: Each angle is 38°.

Your Turn

Vertical angles measure $(4x + 3)°$ and $(x + 39)°$.

Find the measure of each angle.

Your Turn

Vertical angles measure $(4x + 3)°$ and $(x + 39)°$.

Find the measure of each angle.

equal: $4x + 3 = x + 39$

take x off both sides: $3x + 3 = 39$

$3x = 36$, $x = 12$

$4(12) + 3 = 51$

Answer: Each angle is 51°.

Whiteboards · Question 1 of 9

What is the complement of 34°?

Answer it.

Whiteboards · Question 1 of 9

What is the complement of 34°?

$$90 - 34 = 56$$

Answer: 56°

Whiteboards · Question 2 of 9

What is the supplement of 112°?

Answer it.

Whiteboards · Question 2 of 9

What is the supplement of 112°?

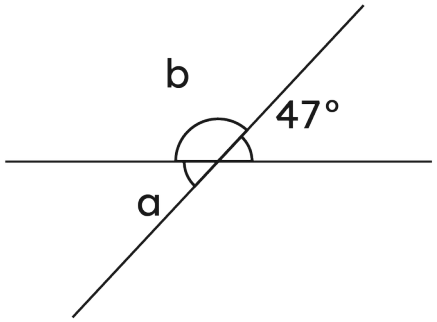
$$180 - 112 = 68$$

Answer: 68°

Whiteboards · Question 3 of 9

Two lines cross. One angle is 47°.

Find a and b.

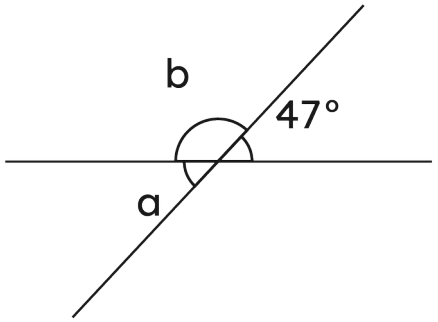


Answer it.

Whiteboards · Question 3 of 9

Two lines cross. One angle is 47°.

Find a and b.



a = 47 (vertical)

b = 180 – 47 = 133 (a linear pair)

Answer: a = 47°, b = 133°

Whiteboards · Question 4 of 9

$\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28°$.

What is $m\angle Q$?

A. 152°

B. 62°

C. 28°

D. 118°

Answer it.

Whiteboards · Question 4 of 9

$\angle P$ and $\angle Q$ are complementary, and $m\angle P = 28°$.

What is $m\angle Q$?

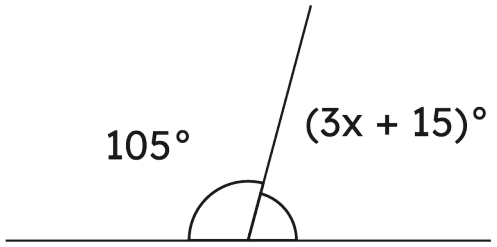
$$90 - 28 = 62$$

Answer: B — 62°

Whiteboards · Question 5 of 9

A linear pair measures $(3x + 15)°$ and $105°$.

Find x.

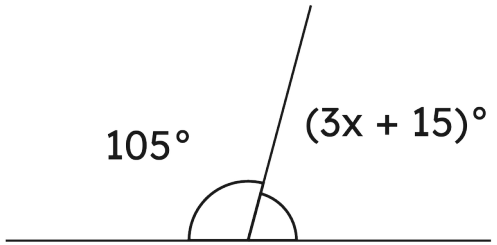


Answer it.

Whiteboards · Question 5 of 9

A linear pair measures $(3x + 15)°$ and $105°$.

Find x.



$3x + 15 + 105 = 180$

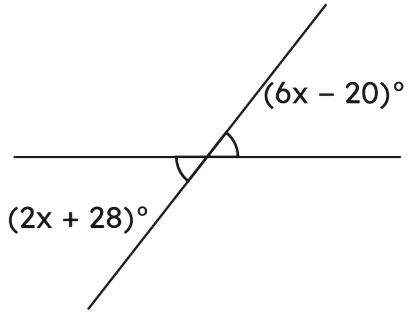
$3x = 60, \quad x = 20$

Answer: $x = 20$

Whiteboards · Question 6 of 9

Vertical angles measure $(6x - 20)°$ and $(2x + 28)°$.

Find the measure of each angle.

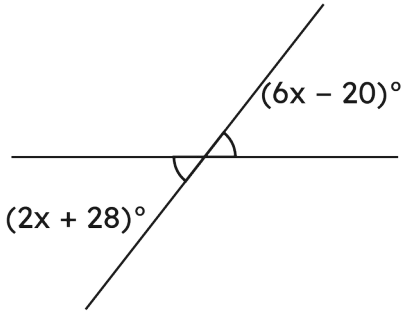


Answer it.

Whiteboards · Question 6 of 9

Vertical angles measure $(6x - 20)°$ and $(2x + 28)°$.

Find the measure of each angle.



equal: $6x - 20 = 2x + 28$, so $4x - 20 = 28$

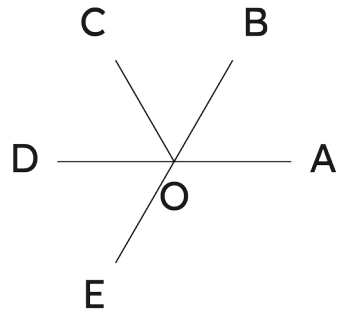
$4x = 48$, $x = 12$

$6(12) - 20 = 52$

Answer: 52°

Whiteboards · Question 7 of 9

Which two angles are adjacent?



A. angles AOB and DOE

B. angles AOB and AOC

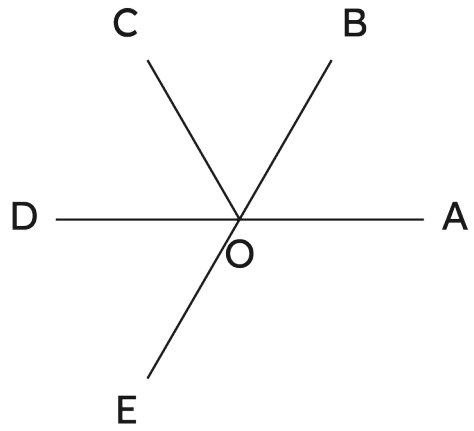
C. angles AOB and BOC

D. angles AOB and COD

Answer it.

Whiteboards · Question 7 of 9

Which two angles are adjacent?



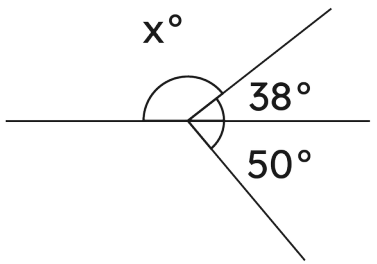
AOB and BOC share the side OB and do not overlap

Answer: C — angles AOB and BOC

Whiteboards · Question 8 of 9

Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°.

Find x.

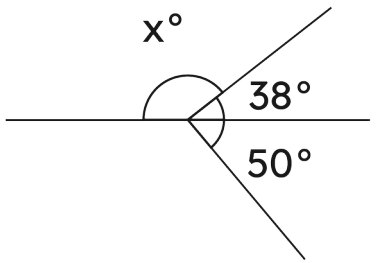


Answer it.

Whiteboards · Question 8 of 9

Above a straight line a ray makes angles of 38° and x°; below it another ray makes an angle of 50°.

Find x.



$38 + x = 180$

$x = 142$

Answer: x = 142

Whiteboards · Question 9 of 9

Can two vertical angles be complementary?

Always, sometimes or never? Explain.

Write your answer in sentences.

Whiteboards · Question 9 of 9

Can two vertical angles be complementary?

Always, sometimes or never? Explain.

vertical: the two are equal

complementary: $90 \div 2 = 45$ each

Answer: Sometimes — only when each is 45°.

Independent Set

1. What is the complement of $61°$?
2. What is the supplement of $76°$?
3. Two lines cross, and one of the four angles is $58°$. Find the other three.
4. A linear pair measures $(5x + 10)°$ and $70°$. Find x and the first angle.
5. Vertical angles measure $(7x - 9)°$ and $(3x + 31)°$. Find the measure of each angle.
6. $\angle PQR$ and $\angle RQS$ share the side QR and together make a straight line, and $m\angle PQR = 65°$. Name the pair and find $m\angle RQS$.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Identify complementary, supplementary, vertical, adjacent, and congruent angles — HGV

Find measures of complementary, supplementary, vertical, and adjacent angles — R2B

Write and solve equations using angle relationships — DBV

GRADE 7 ACCELERATED · UNIT 6 · THREAD B · DAY 2

Interior and Exterior Angles of a Triangle

MA.8.GR.1.5

I can find a missing angle of a triangle, inside or outside, with the Triangle Sum and the Exterior Angle Theorem.

Warm-Up

1. What is the supplement of 83°?
2. How far apart are $(-4, -1)$ and $(4, 5)$?
3. 12 pens cost $9. What does one pen cost?
4. A straight angle is split into 50°, 60° and x°. Find x.

Warm-Up

1. What is the supplement of 83°? **97°**
2. How far apart are $(-4, -1)$ and $(4, 5)$? **10 units**
3. 12 pens cost \$9. What does one pen cost? **\$0.75**
4. A straight angle is split into 50°, 60° and x°. Find x. **70**

Vocabulary

interior angle

an angle inside a triangle, at one of its corners

exterior angle

the angle between one side of a triangle and the next side extended; it makes a linear pair with the interior angle beside it

remote interior angles

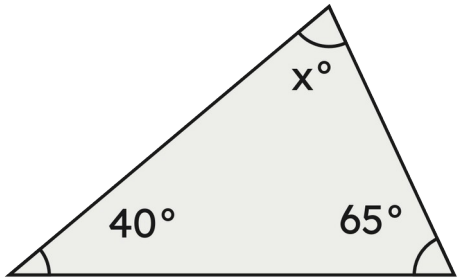
the two interior angles that are not beside a given exterior angle

Triangle Sum Theorem

the three interior angles of a triangle add to 180°

Notes

I. The Triangle Sum: the three angles make 180°

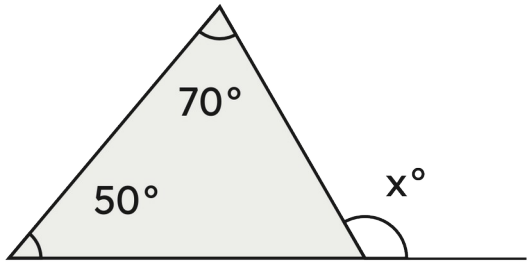


$$x + 65 + 40 = 180, \quad x = 75$$

A triangle's three angles add to 180°: $m\angle A + m\angle B + m\angle C = 180°$

Notes

II. The Exterior Angle Theorem: outside = the two remote angles

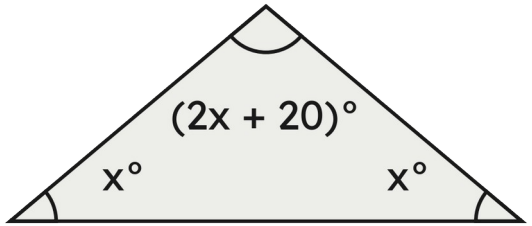


$$x = 50 + 70 = 120 \qquad \text{check:} \quad 60 + 120 = 180$$

Not the angle beside it, and not the angle around the outside.

Notes

III. Angles as expressions

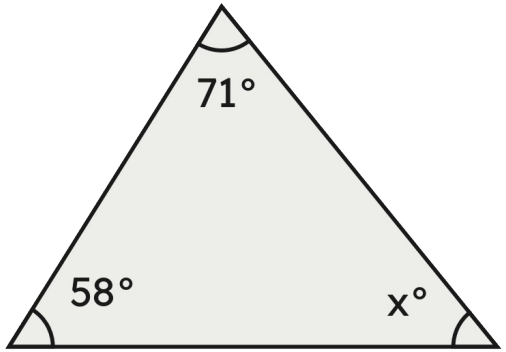


$$x + x + (2x + 20) = 180, \quad 4x = 160, \quad x = 40$$

the angles: 40, 40, 100

Example 1

Two angles of a triangle are $58°$ and $71°$.



Find the third angle.

Example 1

$$58 + 71 = 129$$

$$x = 180 - 129 = 51$$

Answer: The third angle is 51°.

Your Turn

Two angles of a triangle are $39°$ and $84°$.

Find the third angle.

Your Turn

Two angles of a triangle are $39°$ and $84°$.

Find the third angle.

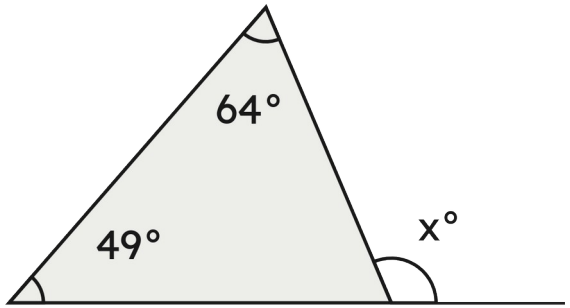
$$39 + 84 = 123$$

$$180 - 123 = 57$$

Answer: $57°$

Example 2

A triangle's remote interior angles are $64°$ and $49°$.



Find the exterior angle x.

Example 2

$x = 64 + 49 = 113$

$180 - 113 = 67, \quad 64 + 49 + 67 = 180$

Answer: The exterior angle is 113°.

Your Turn

A triangle's remote interior angles are $72°$ and $35°$.

Find the exterior angle.

Your Turn

A triangle's remote interior angles are $72°$ and $35°$.

Find the exterior angle.

$$72 + 35 = 107$$

Answer: $107°$

Whiteboards · Question 1 of 9

Two angles of a triangle are 47° and 68°. Find the third angle.

Answer it.

Whiteboards · Question 1 of 9

Two angles of a triangle are 47° and 68°. Find the third angle.

$$47 + 68 = 115$$

$$180 - 115 = 65$$

Answer: 65°

Whiteboards · Question 2 of 9

An isosceles triangle's two base angles are equal. Each is 52°.

Find the third angle.

Answer it.

Whiteboards · Question 2 of 9

An isosceles triangle's two base angles are equal. Each is 52°.

Find the third angle.

$$52 + 52 = 104$$

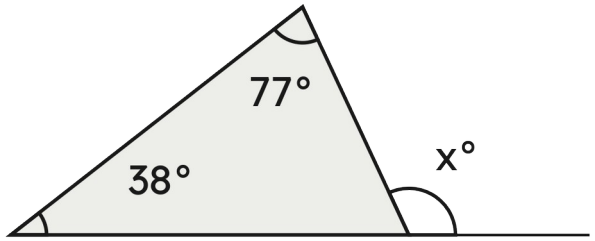
$$180 - 104 = 76$$

Answer: 76°

Whiteboards · Question 3 of 9

A triangle's remote interior angles are 38° and 77°.

Find the exterior angle x.

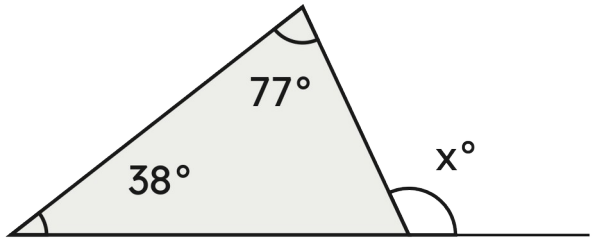


Answer it.

Whiteboards · Question 3 of 9

A triangle's remote interior angles are 38° and 77°.

Find the exterior angle x.



$x = 38 + 77 = 115$

Answer: 115°

Whiteboards · Question 4 of 9

A triangle's angles are 45°, 60° and 75°.

What is the exterior angle beside the 75°?

A. 75°

B. 105°

C. 285°

D. 60°

Answer it.

Whiteboards · Question 4 of 9

A triangle's angles are 45°, 60° and 75°.

What is the exterior angle beside the 75°?

$$45 + 60 = 105$$

check: $105 + 75 = 180$

Answer: B — 105°

Whiteboards · Question 5 of 9

An exterior angle of a triangle is 130°. One remote interior angle is 55°.

Find the other remote interior angle.

Answer it.

Whiteboards · Question 5 of 9

An exterior angle of a triangle is 130°. One remote interior angle is 55°.

Find the other remote interior angle.

$$55 + x = 130$$

$$x = 75$$

Answer: 75°

Whiteboards · Question 6 of 9

A triangle's angles are $x°$, $3x°$ and $(2x + 12)°$.

Find each angle.

Answer it.

Whiteboards · Question 6 of 9

A triangle's angles are $x°$, $3x°$ and $(2x + 12)°$.

Find each angle.

$$x + 3x + 2x + 12 = 180, \quad 6x = 168, \quad x = 28$$

$$3(28) = 84, \quad 2(28) + 12 = 68$$

Answer: 28°, 84° and 68°

Whiteboards · Question 7 of 9

An exterior angle measures $(5x + 4)°$.

Its remote interior angles measure $52°$ and $(3x)°$.

Which equation finds x?

A. $5x + 4 + 3x = 180$

B. $5x + 4 = 3x$

C. $5x + 4 = 52 + 3x$

D. $5x + 4 + 52 + 3x = 180$

Answer it.

Whiteboards · Question 7 of 9

An exterior angle measures $(5x + 4)°$.

Its remote interior angles measure $52°$ and $(3x)°$.

Which equation finds x?

outside = the two far ones: $5x + 4 = 52 + 3x$

$2x = 48$, $x = 24$

Answer: C — $5x + 4 = 52 + 3x$

Whiteboards · Question 8 of 9

A triangle has angles of 41° and 66°, and its longest side is 9 cm.

Find its third angle.

Answer it.

Whiteboards · Question 8 of 9

A triangle has angles of 41° and 66°, and its longest side is 9 cm.

Find its third angle.

$$41 + 66 = 107$$

$$180 - 107 = 73$$

Answer: 73°

Whiteboards · Question 9 of 9

Why is an exterior angle always bigger than either remote interior angle?

Write your answer in sentences.

Whiteboards · Question 9 of 9

Why is an exterior angle always bigger than either remote interior angle?

for example: $40 + 65 = 105$, more than 40 and more than 65

Answer: It is the two remote angles added, each more than 0°.

Independent Set

1. Two angles of a triangle are $63°$ and $79°$. Find the third angle.
2. An isosceles triangle's two base angles are equal. The angle between its two equal sides is $40°$. Find each base angle.
3. A triangle's remote interior angles are $46°$ and $58°$. Find the exterior angle.
4. An exterior angle of a triangle is $140°$. One remote interior angle is $62°$. Find the other.
5. A triangle's angles are $(2x)°$, $(3x)°$ and $(4x)°$. Find each angle.
6. An exterior angle of a triangle measures $(4x + 10)°$; its remote interior angles measure $50°$ and $(2x)°$. Write an equation and find x.

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Find missing angles in triangles — JFJ

Exterior Angle Theorem — FMP

Triangle Angle-Sum Theorem — 6Q6